

Online Appendix to “Scalable Just-in-Time Price Elasticity Estimation”

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A Identification Derivations

A.1 Derivation of Equation (3)

Using the smoothness conditions stated in Assumptions 3 and 4, we take derivatives on both sides of the reduced-form equation $h_j = f_j + \lambda$ (Section 2) with respect to prices and instruments. By the chain rule, we get

$$\underbrace{\partial_{\mathbf{p}_t} h_j(\mathbf{p}_t, \mathbf{x}_t, \tilde{\mathbf{z}}_t)}_{J \times 1} = \underbrace{\partial_{\mathbf{p}_t} f_j(\mathbf{p}_t, \mathbf{x}_t)}_{J \times 1} + \underbrace{\partial_{\mathbf{u}_t} \lambda(\mathbf{u}_t)}_{J \times 1},$$

$$\underbrace{\partial_{\tilde{\mathbf{z}}_t} h_j(\mathbf{p}_t, \mathbf{x}_t, \tilde{\mathbf{z}}_t)}_{d_{\tilde{\mathbf{z}}} \times 1} = - \underbrace{\partial_{\tilde{\mathbf{z}}_t} \mathbf{g}(\mathbf{z}_t)}_{d_{\tilde{\mathbf{z}}} \times J} \underbrace{\partial_{\mathbf{u}_t} \lambda(\mathbf{u}_t)}_{J \times 1}.$$

Here $\partial_{\mathbf{p}_t} h_j(\mathbf{p}_t, \mathbf{x}_t, \tilde{\mathbf{z}}_t)$ is the Jacobian matrix of conditional demand function h_j with respect to price vector $\mathbf{p}_t$, evaluated at point $(\mathbf{p}_t, \mathbf{x}_t, \tilde{\mathbf{z}}_t)$. The other terms are defined in the same fashion. We explicitly list the dimensions of these Jacobian matrices underneath to avoid potential confusion on various definitions of the Jacobian matrix.

We can rearrange the above two equations and get

$$\underbrace{\partial_{\tilde{\mathbf{z}}_t} \mathbf{g}(\mathbf{z}_t)}_{d_{\tilde{\mathbf{z}}} \times J} \underbrace{\partial_{\mathbf{p}_t} f_j(\mathbf{p}_t, \mathbf{x}_t)}_{J \times 1} = \underbrace{\partial_{\tilde{\mathbf{z}}_t} h_j(\mathbf{p}_t, \mathbf{x}_t, \tilde{\mathbf{z}}_t)}_{d_{\tilde{\mathbf{z}}} \times 1} + \underbrace{\partial_{\tilde{\mathbf{z}}_t} \mathbf{g}(\mathbf{z}_t)}_{d_{\tilde{\mathbf{z}}} \times J} \underbrace{\partial_{\mathbf{p}_t} h_j(\mathbf{p}_t, \mathbf{x}_t, \tilde{\mathbf{z}}_t)}_{J \times 1},$$

which gives us a system of $d_{\tilde{\mathbf{z}}}$ linear equations in J unknowns, whose solution is to be discussed in the following two scenarios.

When $d_{\tilde{\mathbf{z}}} > J$, that is, when the number of excluded instrumental variables is larger than the number of endogenous prices, it is an over-identified system. Since $\partial_{\tilde{\mathbf{z}}_t} \mathbf{g}(\tilde{\mathbf{z}}_t)$ has full column rank, we are able to obtain a minimum distance solution,

$$\underbrace{\partial_{\mathbf{p}_t} f_j(\mathbf{p}_t, \mathbf{x}_t)}_{J \times 1} = \underbrace{\partial_{\mathbf{p}_t} h_j(\mathbf{p}_t, \mathbf{x}_t, \tilde{\mathbf{z}}_t)}_{J \times 1} + \underbrace{(\partial_{\tilde{\mathbf{z}}_t} \mathbf{g}(\mathbf{z}_t))^\top}_{J \times d_{\tilde{\mathbf{z}}}} \underbrace{\partial_{\tilde{\mathbf{z}}_t} \mathbf{g}(\mathbf{z}_t)}_{d_{\tilde{\mathbf{z}}} \times J}^{-1} \underbrace{\partial_{\tilde{\mathbf{z}}_t} \mathbf{g}(\mathbf{z}_t)}_{J \times d_{\tilde{\mathbf{z}}}}^\top \underbrace{\partial_{\tilde{\mathbf{z}}_t} h_j(\mathbf{p}_t, \mathbf{x}_t, \tilde{\mathbf{z}}_t)}_{d_{\tilde{\mathbf{z}}} \times 1}.$$

A.2 Derivation of special cases

Case 1 When there are J endogenous prices, for each of the endogeneous price p_i there is exactly one excluded instrumental variable $\tilde{z}_i$, we get

$$\begin{pmatrix} \frac{\partial f_j(\mathbf{p}_t, \mathbf{x}_t)}{\partial p_{1t}} \\ \vdots \\ \frac{\partial f_j(\mathbf{p}_t, \mathbf{x}_t)}{\partial p_{Jt}} \end{pmatrix} = \begin{pmatrix} \frac{\partial h_j(\mathbf{p}_t, \mathbf{x}_t)}{\partial p_{1t}} \\ \vdots \\ \frac{\partial h_j(\mathbf{p}_t, \mathbf{x}_t)}{\partial p_{Jt}} \end{pmatrix} + \begin{pmatrix} \left(\frac{\partial g_1(z_{1t})}{\partial \tilde{z}_{1t}}\right)^{-1} & & \\ & \ddots & \\ & & \left(\frac{\partial g_J(z_{Jt})}{\partial \tilde{z}_{Jt}}\right)^{-1} \end{pmatrix} \begin{pmatrix} \frac{\partial h_j(\mathbf{p}_t, \mathbf{x}_t)}{\partial \tilde{z}_{1t}} \\ \vdots \\ \frac{\partial h_j(\mathbf{p}_t, \mathbf{x}_t)}{\partial \tilde{z}_{Jt}} \end{pmatrix}.$$

In this special case, it holds that, for $i = 1, 2, \dots, J$,

$$\frac{\partial f_j(\mathbf{p}_t, \mathbf{x}_t)}{\partial p_{it}} = \left(\frac{\partial g_i(z_{it})}{\partial \tilde{z}_{it}}\right)^{-1} \frac{\partial h_j(\mathbf{p}_t, \mathbf{x}_t)}{\partial \tilde{z}_{it}}.$$

Case 2 When there are one endogenous price p_t , for this endogenous price there are instrumental variables $\tilde{z}_{1t}$ and $\tilde{z}_{2t}$, we get

$$\frac{\partial f_j(p_t, \mathbf{x}_t)}{\partial p_t} = \frac{\partial h_j(p_t, \mathbf{x}_t)}{\partial p_t} + \left(\left(\left(\frac{\partial g(\mathbf{z}_t)}{\partial \tilde{z}_{1t}} \right)^2 + \left(\frac{\partial g(\mathbf{z}_t)}{\partial \tilde{z}_{2t}} \right)^2 \right)^{-1} \begin{pmatrix} \frac{\partial g(\mathbf{z}_t)}{\partial \tilde{z}_{1t}} & \frac{\partial g(\mathbf{z}_t)}{\partial \tilde{z}_{2t}} \end{pmatrix} \begin{pmatrix} \frac{\partial h_j(\mathbf{p}_t, \mathbf{x}_t)}{\partial \tilde{z}_{1t}} \\ \frac{\partial h_j(\mathbf{p}_t, \mathbf{x}_t)}{\partial \tilde{z}_{2t}} \end{pmatrix} \right).$$

In this special case, the price effect is a weighted average using both instrumental variables.

Case 3 When there are two endogenous prices p_{1t} and p_{2t} , for p_{1t} there are two excluded instruments $\tilde{z}_{1t}$ and $\tilde{z}_{2t}$, for p_{2t} there are one excluded instrument $\tilde{z}_{3t}$, we get

$$\begin{pmatrix} \frac{\partial f_j(\mathbf{p}_t, \mathbf{x}_t)}{\partial p_{1t}} \\ \frac{\partial f_j(\mathbf{p}_t, \mathbf{x}_t)}{\partial p_{2t}} \end{pmatrix} = \begin{pmatrix} \frac{\partial h_j(\mathbf{p}_t, \mathbf{x}_t)}{\partial p_{1t}} \\ \frac{\partial h_j(\mathbf{p}_t, \mathbf{x}_t)}{\partial p_{2t}} \end{pmatrix} + \begin{pmatrix} \left(\left(\frac{\partial g_1(z_{1t}, z_{2t})}{\partial \tilde{z}_{1t}} \right)^2 + \left(\frac{\partial g_1(z_{1t}, z_{2t})}{\partial \tilde{z}_{2t}} \right)^2 \right)^{-1} & \\ & \left(\frac{\partial g_2(z_{3t})}{\partial \tilde{z}_{3t}} \right)^{-2} \end{pmatrix} \begin{pmatrix} \frac{\partial g_1(z_{1t}, z_{2t})}{\partial \tilde{z}_{1t}} & \frac{\partial g_1(z_{1t}, z_{2t})}{\partial \tilde{z}_{2t}} \\ \frac{\partial g_2(z_{3t})}{\partial \tilde{z}_{3t}} \end{pmatrix} \begin{pmatrix} \frac{\partial h_j(\mathbf{p}_t, \mathbf{x}_t)}{\partial \tilde{z}_{1t}} \\ \frac{\partial h_j(\mathbf{p}_t, \mathbf{x}_t)}{\partial \tilde{z}_{2t}} \\ \frac{\partial h_j(\mathbf{p}_t, \mathbf{x}_t)}{\partial \tilde{z}_{3t}} \end{pmatrix}.$$

In this special case, it holds that,

$$\frac{\partial f_j(p_t, \mathbf{x}_t)}{\partial p_{1t}} = \frac{\partial h_j(p_t, \mathbf{x}_t)}{\partial p_{1t}} + \left(\left(\left(\frac{\partial g_1(z_{1t}, z_{2t})}{\partial \tilde{z}_{1t}} \right)^2 + \left(\frac{\partial g_1(z_{1t}, z_{2t})}{\partial \tilde{z}_{2t}} \right)^2 \right)^{-1} \begin{pmatrix} \frac{\partial g_1(z_{1t}, z_{2t})}{\partial \tilde{z}_{1t}} & \frac{\partial g_1(z_{1t}, z_{2t})}{\partial \tilde{z}_{2t}} \end{pmatrix} \begin{pmatrix} \frac{\partial h_j(\mathbf{p}_t, \mathbf{x}_t)}{\partial \tilde{z}_{1t}} \\ \frac{\partial h_j(\mathbf{p}_t, \mathbf{x}_t)}{\partial \tilde{z}_{2t}} \end{pmatrix} \right),$$

$$\frac{\partial f_j(\mathbf{p}_t, \mathbf{x}_t)}{\partial p_{2t}} = \left(\frac{\partial g_2(z_{3t})}{\partial \tilde{z}_{3t}} \right)^{-1} \frac{\partial h_j(\mathbf{p}_t, \mathbf{x}_t)}{\partial \tilde{z}_{3t}}.$$

Case 4 When there are two endogenous prices p_{1t} and p_{2t} , for p_{1t} there are one excluded instruments $\tilde{z}_{1t}$, for p_{2t} there are one excluded instrument $\tilde{z}_{2t}$, for both of them there is a common excluded instrument $\tilde{z}_{3t}$, we get

$$\begin{pmatrix} \frac{\partial f_j(\mathbf{p}_t, \mathbf{x}_t)}{\partial p_{1t}} \\ \frac{\partial f_j(\mathbf{p}_t, \mathbf{x}_t)}{\partial p_{2t}} \end{pmatrix} = \begin{pmatrix} \frac{\partial h_j(\mathbf{p}_t, \mathbf{x}_t)}{\partial p_{1t}} \\ \frac{\partial h_j(\mathbf{p}_t, \mathbf{x}_t)}{\partial p_{2t}} \end{pmatrix} + \begin{pmatrix} \left(\frac{\partial g_1(z_{1t}, z_{3t})}{\partial \tilde{z}_{1t}} \right)^2 + \left(\frac{\partial g_1(z_{1t}, z_{3t})}{\partial \tilde{z}_{3t}} \right)^2 & \frac{\partial g_1(z_{1t}, z_{3t})}{\partial \tilde{z}_{3t}} \frac{\partial g_2(z_{2t}, z_{3t})}{\partial \tilde{z}_{3t}} \\ \frac{\partial g_2(z_{2t}, z_{3t})}{\partial \tilde{z}_{3t}} \frac{\partial g_1(z_{1t}, z_{3t})}{\partial \tilde{z}_{3t}} & \left(\frac{\partial g_2(z_{2t}, z_{3t})}{\partial \tilde{z}_{2t}} \right)^2 + \left(\frac{\partial g_2(z_{2t}, z_{3t})}{\partial \tilde{z}_{3t}} \right)^2 \end{pmatrix}^{-1} \begin{pmatrix} \frac{\partial g_1(z_{1t}, z_{3t})}{\partial \tilde{z}_{1t}} & \frac{\partial g_1(z_{1t}, z_{3t})}{\partial \tilde{z}_{3t}} \\ \frac{\partial g_2(z_{2t}, z_{3t})}{\partial \tilde{z}_{2t}} & \frac{\partial g_2(z_{2t}, z_{3t})}{\partial \tilde{z}_{3t}} \end{pmatrix} \begin{pmatrix} \frac{\partial h_j(\mathbf{p}_t, \mathbf{x}_t)}{\partial \tilde{z}_{1t}} \\ \frac{\partial h_j(\mathbf{p}_t, \mathbf{x}_t)}{\partial \tilde{z}_{2t}} \\ \frac{\partial h_j(\mathbf{p}_t, \mathbf{x}_t)}{\partial \tilde{z}_{3t}} \end{pmatrix}.$$

A.3 A graphical illustration of the identification results

Our central identification result in equation (3) can be illustrated by a directed acyclic graph (Pearl, 2009). For ease of exposition, we further simplify equation (3) to the case with one endogenous price and one excluded instrument, i.e.,

$$\partial_p f(p_t, \mathbf{x}_t) = \partial_p h(p_t, \mathbf{x}_t, \tilde{z}_t) - [-\partial_{\tilde{z}_t} g(z_t)^{-1} \partial_{\tilde{z}_t} h(p_t, \mathbf{x}_t, \tilde{z}_t)].$$

With this simplification, we can visualize our identification result in Figure A1. Our goal is to evaluate the causal effect of price on sales quantity, $\partial_p f(p_t, \mathbf{x}_t)$, which is channel ② in the figure. This partial effect is a causal effect because the unobserved confounder is held fixed. However, the correlation between sales quantity and prices in the observational data, or the “total effect,” is the sum of the causal effect (channel ②) and selection bias (channels ③ and ④).

Under our identifying assumptions, we can back out how changes in the instrumental variable relate to changes in price ($\partial_z g(z_t)$, or channel ①). We can also identify $\partial_z h(p_t, \mathbf{x}_t, \tilde{z}_t)$, which is how the instrumental variable is correlated with sales quantity while holding the price constant (channel ① $\rightarrow$ ③ $\leftarrow$ ④). The indirect effect of price on sales quantity (channel ③ $\rightarrow$ ④) can be worked out by using $\partial_p h(p_t, \mathbf{x}_t, \tilde{z}_t)$ to divide $\partial_{\tilde{z}_t} g(z_t)^{-1}$ and then flipping the sign. When this is done, $\partial_{\mathbf{p}_t} f_j(\mathbf{p}_t, \mathbf{x}_t)$ is identified.

Figure A1: Graphical interpretation of the identification result



Note: This figure depicts a simplified relationship between sales quantity, price, an unobserved confounder, and instrumental variables, using directed acyclic graphs (Pearl, 2009).

B Algorithms

Algorithm 1 Algorithm for Debiased Bagged Nearest Neighbors

Input: Sample $(y_i, \mathbf{w}_i)_{i=1}^n$; evaluation point $\mathbf{w}_0$; tuning parameter m

Output: Debiased bagged nearest neighbors estimate $\hat{\mu}(\mathbf{w}_0)$

procedure DEBIASEDBNN($((y_i, \mathbf{w}_i)_{i=1}^n, \mathbf{w}_0, m)$)

 Calculate Euclidean distances $\|\mathbf{w}_i - \mathbf{w}_0\|$ for all i

 Sort y_i based on distances $\|\mathbf{w}_i - \mathbf{w}_0\|$ in ascending order to get $y_{(1)}, \dots, y_{(n)}$

 Compute base weights $\mathbf{w}^{(1)}$ for subsample size m using Equation (6)

 Compute auxiliary weights $\mathbf{w}^{(2)}$ for subsample size $2m$ using Equation (6)

 Compute generalized jackknife: $\mathbf{w}^* = -1.70 \cdot \mathbf{w}^{(1)} + 2.70 \cdot \mathbf{w}^{(2)}$

return Dot product $\mathbf{w}^* \cdot \mathbf{y}_{\text{sorted}}$

end procedure

Note: The jackknife weights -1.70 and 2.70 are derived for dimension $d = 3$. For other dimensions, recompute following Footnote 5. The selection of m is discussed in Section 4.4.

Algorithm 2 Nonparametric Price Elasticity Estimation

1: **Input:** Data $(s_{jt}, \mathbf{p}_t, \mathbf{x}_t, \tilde{\mathbf{z}}_t)_{t=1}^n$; evaluation point $(\mathbf{p}_0, \mathbf{x}_0, \tilde{\mathbf{z}}_0)$; tuning parameters m, h_n

2: **Step 1:** Estimate $\hat{h}_j(\mathbf{p}_0, \mathbf{x}_0, \tilde{\mathbf{z}}_0) = \tau_n^{(m)}(\mathbf{p}_0, \mathbf{x}_0, \tilde{\mathbf{z}}_0)$ using Algorithm 1

3: Estimate price equation $g(\mathbf{z}_0)$ parametrically or via $\hat{g}(\mathbf{z}_0) = \tau_n^{(m)}(\mathbf{z}_0)$

4: **Step 2:** Compute forward finite differences with bandwidth h_n :

5: $\widehat{\partial_{p_k} h_j} = [\hat{h}_j(\mathbf{p}_0 + h_n \mathbf{e}_k, \mathbf{x}_0, \tilde{\mathbf{z}}_0) - \hat{h}_j(\mathbf{p}_0, \mathbf{x}_0, \tilde{\mathbf{z}}_0)] / h_n$

6: $\widehat{\partial_{z_l} h_j} = [\hat{h}_j(\mathbf{p}_0, \mathbf{x}_0, \tilde{\mathbf{z}}_0 + h_n \mathbf{e}_l) - \hat{h}_j(\mathbf{p}_0, \mathbf{x}_0, \tilde{\mathbf{z}}_0)] / h_n$

7: $\widehat{\partial_{z_l} g} = [\hat{g}(\mathbf{z}_0 + h_n \mathbf{e}_l) - \hat{g}(\mathbf{z}_0)] / h_n$

8: **Step 3:** Structural recovery: $\widehat{\partial_{p_k} f_j} = \widehat{\partial_{p_k} h_j} + (\widehat{\partial_{z_l} g})^{-1} \widehat{\partial_{z_l} h_j}$

9: **Step 4:** Elasticity: $\hat{\varepsilon}_{jk}(\mathbf{p}_0, \mathbf{x}_0) = \widehat{\partial_{p_k} f_j} \cdot p_{k0} / \hat{h}_j(\mathbf{p}_0, \mathbf{x}_0, \tilde{\mathbf{z}}_0)$

10: **Output:** Price elasticity estimate $\hat{\varepsilon}_{jk}(\mathbf{p}_0, \mathbf{x}_0)$

C Theoretical Properties of Bagged Nearest Neighbors

The bagged nearest neighbors estimator is a nonparametric tool for conditional expectation estimation. This appendix presents the theoretical properties of bagged nearest neighbors, including asymptotic normality, uniform convergence, and bootstrap validity. For completeness and clarity, we begin with the model setup, followed by known properties, and conclude with new uniform convergence and bootstrap results.

C.1 Model Setting and Assumptions

Consider a sample of independent and identically distributed (i.i.d.) observations $\{(\mathbf{w}_i, y_i)\}_{i=1}^n$ from the following nonparametric model:

$$y = \mu(\mathbf{w}) + \epsilon,$$

where y is the response, $\mathbf{w} \in \mathbb{R}^d$ represents the vector of covariates with fixed dimensionality d , $\mu(\mathbf{w})$ is the unknown true regression function, and ϵ is the random disturbance.

Definition The bagged nearest neighbors estimator with subsampling size m at $\mathbf{w}_0$ is:

$$\tau_n(m)(\mathbf{w}_0) = \binom{n}{m}^{-1} \sum_{1 \leq i_1 < \dots < i_m \leq n} \Phi(\mathbf{w}_0; Z_{i_1}, \dots, Z_{i_m}),$$

where $\Phi(\mathbf{w}_0; Z_{i_1}, \dots, Z_{i_m}) = y_{(1)}(Z_{i_1}, \dots, Z_{i_m})$ is the response of the nearest neighbor in Euclidean distance to $\mathbf{w}_0$ in the subsample $\{Z_{i_1}, \dots, Z_{i_m}\}$, and $Z_i = (\mathbf{w}_i, y_i)$.

Assumption C.1. We have an i.i.d. sample $\{(\mathbf{w}_1, y_1), (\mathbf{w}_2, y_2), \dots, (\mathbf{w}_n, y_n)\}$.

Assumption C.2. The support of $\mathbf{w}$, denoted as $\text{supp}(\mathbf{w})$, is compact.

Assumption C.3. The density $\nu(\cdot)$ of $\mathbf{w}$ is bounded away from 0 and ∞ , $\nu(\cdot)$ and $\mu(\cdot)$ are both twice continuously differentiable with bounded second derivatives.

Assumption C.4. The kernel Φ is bounded, i.e., there exists a constant $B > 0$ such that $|\Phi| \leq B$, and Φ has finite second moment, $\mathbb{E} \Phi^2 < \infty$. ϵ is independent of $\mathbf{w}$, has zero mean and finite variance $\sigma_\epsilon^2 > 0$.

Assumption C.5. As $n \rightarrow \infty$, we have $m \rightarrow \infty$ and $m/n \rightarrow 0$.

C.2 Established Asymptotic Properties

Theorem C.1. *Given $\mathbf{w}_0 \in \text{supp}(\mathbf{w})$, $d \geq 2$, under Assumptions C.1–C.5,*

$$\mathbb{E} \tau_n(m)(\mathbf{w}_0) = \mu(\mathbf{w}_0) + B(m),$$

$$B(m) = \Gamma(2/d + 1) \frac{\nu(\mathbf{w}_0) \text{tr}(\mu''(\mathbf{w}_0)) + 2 \mu'(\mathbf{w}_0)^\top \nu'(\mathbf{w}_0)}{2 d V_d^{2/d} \nu(\mathbf{w}_0)^{1+2/d}} m^{-2/d} + O(m^{-4/d}),$$

where $V_d = \frac{\pi^{d/2}}{\Gamma(1+d/2)}$, $\Gamma(\cdot)$ denotes the Gamma function, $\nu'(\mathbf{w}_0)$ and $\mu'(\mathbf{w}_0)$ are the first order gradients at $\mathbf{w}_0$ for $\nu(\mathbf{w})$ and $\mu(\mathbf{w})$, respectively, $\mu''(\mathbf{w}_0)$ is the Hessian matrix of $\mu(\cdot)$ at $\mathbf{w}_0$, and $\text{tr}(\cdot)$ gives the trace.

Theorem C.2. *Given $\mathbf{w}_0 \in \text{supp}(\mathbf{w})$, under Assumptions C.1–C.5, then for some positive σ_n with $\sigma_n^2(\mathbf{w}_0) = O(m/n)$, as $n \rightarrow \infty$,*

$$\frac{\tau_n(m)(\mathbf{w}_0) - \mu(\mathbf{w}_0) - B(m)}{\sigma_n(\mathbf{w}_0)} \xrightarrow{\mathcal{D}} N(0, 1).$$

The point-wise asymptotic properties of the bagged nearest neighbors estimator were established in Demirkaya et al. (2022). Building upon their work, we present several new theoretical results in this appendix. Our results extend the point-wise properties to uniform convergence, derivative and quotient estimation, and uniform bootstrap validity. For readers interested in the mathematical details, we provide comprehensive proofs of these theorems and their supporting lemmas in Section C.6.

C.3 Uniform Convergence

In the following theorems, we establish the uniform convergence of the bagged nearest neighbors estimator. These results strengthen the point-wise properties to hold uniformly across the support (under the supreme norm). The key insight enabling these theorems is to show that the bagged nearest neighbors estimator is “well-behaved,” that is, the estimator’s fluctuations can be uniformly controlled. This uniform convergence is crucial and provides a solid foundation for our more complex statistical procedures.

Theorem C.3 (Uniform Convergence). *Under Assumptions C.1–C.5, the bagged nearest neighbors estimator converges uniformly in probability to the true regression function. Specif-*

ically, for every $\epsilon > 0$,

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(\sup_{\mathbf{w}_0 \in \text{supp}(\mathbf{w})} |\tau_n(m)(\mathbf{w}_0) - \mu(\mathbf{w}_0)| > \epsilon \right) = 0,$$

where $m \rightarrow \infty$ and $m/n \rightarrow 0$ as $n \rightarrow \infty$.

We present two proofs of the above theorem in Section C.6. The first proof utilizes the U-statistic representation of bagged nearest neighbors and builds on a modern concentration inequality for U-statistics. The second proof is a more technical approach that employs the concepts of stochastic equicontinuity from Newey (1991). Both proof strategies demonstrate that the estimator is well-behaved in fluctuations across its support.

Theorem C.4 (Uniform Asymptotic Normality). *Suppose that Assumptions C.1–C.5 hold. Then, as $n \rightarrow \infty$, $m \rightarrow \infty$, and $m/n \rightarrow 0$, it holds that*

$$\sup_{\mathbf{w} \in \text{supp}(\mathbf{w})} \sup_{x \in \mathbb{R}} \left| \mathbb{P} \left(\frac{\tau_n(m)(\mathbf{w}) - \mu(\mathbf{w}) - B(m)}{\sigma_n(\mathbf{w})} \leq x \right) - \Phi(x) \right| \rightarrow 0,$$

where $\tau_n(m)(\mathbf{w})$ is the bagged nearest neighbors estimator of the true regression function $\mu(\mathbf{w})$, $B(m)$ is the bias term as defined in Theorem C.1, $\sup_{\mathbf{w}} \sigma_n^2(\mathbf{w}) = O(m/n)$ and $\inf_{\mathbf{w}} \sigma^2(\mathbf{w}) \geq c > 0$, and here $\Phi(x)$ is the cumulative distribution function of the standard normal distribution.

This result shows that the point-wise asymptotic normality result in Theorem C.2 holds uniformly. Its proof uses Hoeffding’s decomposition of U-statistics and establishes uniform asymptotic normality by verifying the conditions of the bracketing central limit theorem. Indeed, these bracketing conditions serve to control the fluctuations of the bagged nearest neighbors estimator, providing another perspective on its well-behaved nature.

C.4 Uniform Convergence of Derivative and Quotient Estimation

The following theorems establish the theoretical foundations of our estimation approach. We first prove the uniform consistency of partial derivative estimation using finite differences of bagged nearest neighbors estimators. We then demonstrate the uniform consistency of elasticity-type estimators constructed as quotients of bagged nearest neighbors estimators. These uniform consistency results ensure the reliability of both estimation procedures across their entire support. Additionally, we provide theoretical guidance for selecting the optimal bandwidth in the finite difference scheme.

Theorem C.5 (Uniform Consistency of Derivative Estimation using BNN). *Let $\mathbf{e}_j$ be the j th standard basis vector in $\mathbb{R}^d$. Under Assumptions C.1–C.5, and assuming that $\mu(\mathbf{w})$ has continuous third-order partial derivatives, let $h_n > 0$ be a bandwidth sequence such that $h_n \rightarrow 0$ and $m^{4/d}h_n \rightarrow \infty$ as $n \rightarrow \infty$, $m \rightarrow \infty$, and $m/n \rightarrow 0$. Then the finite difference estimator*

$$\Delta_{h_n}\tau_n(m)(\mathbf{w}) = \frac{\tau_n(m)(\mathbf{w} + h_n\mathbf{e}_j) - \tau_n(m)(\mathbf{w})}{h_n}$$

is uniformly consistent for $\partial\mu(\mathbf{w})/\partial w_j$ over $\text{supp}(\mathbf{w})$. Specifically, for any $\epsilon > 0$,

$$\mathbb{P}\left(\sup_{\mathbf{w} \in \text{supp}(\mathbf{w})} |\Delta_{h_n}\tau_n(m)(\mathbf{w}) - \frac{\partial\mu}{\partial w_j}(\mathbf{w})| > \epsilon\right) \rightarrow 0.$$

When using the debiased BNN estimator to construct the finite difference, the same result holds with the bandwidth conditions $h_n \rightarrow 0$ and $m^{6/d}h_n \rightarrow \infty$.

Theorem C.6 (Rate for Bandwidth/Stepsize Selection). *Under Assumptions C.1 – C.5, and assuming that $\mu(\mathbf{w})$ has continuous third-order partial derivatives, the optimal rates for h_n and m that minimize the mean squared error (MSE) of the derivative estimator $\Delta_h\tau_n(m)(\mathbf{w})$ defined in Theorem C.5 uniformly over $\text{supp}(\mathbf{w})$ are:*

$$h = O(n^{-\frac{2}{d+8}}), \quad m = O(n^{\frac{d}{d+8}}),$$

where d is the dimension of $\mathbf{w}$. Moreover, if we use debiased BNN estimator to construct the finite difference, the optimal rates are

$$h = O(n^{-\frac{3}{d+12}}), \quad m = O(n^{\frac{d}{d+12}}).$$

Theorem C.7 (Uniform Consistency of Quotient Estimator). *Let $\tau_{n_1}^{(1)}(m_1)(\mathbf{w})$ and $\tau_{n_2}^{(2)}(m_2)(\mathbf{w})$ be bagged nearest neighbors estimators for functions $\mu_1(\mathbf{w})$ and $\mu_2(\mathbf{w})$ respectively, where n_1, n_2 are the respective sample sizes and m_1, m_2 are the respective subsampling sizes. Assume that Assumptions C.1–C.5 hold for both estimators. Further assume that $|\mu_2(\mathbf{w})|$ is bounded away from zero, i.e., there exists $c > 0$ such that $|\mu_2(\mathbf{w})| > c$ for all $\mathbf{w} \in \text{supp}(\mathbf{w})$. As $\min(n_1, n_2) \rightarrow \infty$, we have $m_1, m_2 \rightarrow \infty$ and $m_1/n_1, m_2/n_2 \rightarrow 0$. Define the quotient estimator:*

$$\hat{\theta}_n(\mathbf{w}) = \frac{\tau_{n_1}^{(1)}(m_1)(\mathbf{w})}{\tau_{n_2}^{(2)}(m_2)(\mathbf{w})}.$$

Then $\hat{\theta}_n(\mathbf{w})$ is uniformly consistent for $\theta(\mathbf{w}) = \mu_1(\mathbf{w})/\mu_2(\mathbf{w})$, i.e.,

$$\sup_{\mathbf{w} \in \text{supp}(\mathbf{w})} |\hat{\theta}_n(\mathbf{w}) - \theta(\mathbf{w})| \xrightarrow{p} 0.$$

These results for finite differences and quotient transformations are special cases of a more general continuous mapping principle, presented in the following Lemma. This principle extends beyond elasticity estimation to provide theoretical justification for a broad range of nonlinear transformations of the estimators. The proof proceeds by first establishing the result for two sequences and then generalizing to the multiple sequence case.

Lemma 1 (Uniform Continuous Mapping Theorem for Multiple Sequences). *Let $X_n^{(i)}(\mathbf{w})$, for $i = 1, \dots, k$, be k sequences of random processes indexed by $\mathbf{w} \in \text{supp}(\mathbf{w})$, where $\text{supp}(\mathbf{w})$ is compact. Let $g : \mathbb{R}^k \rightarrow \mathbb{R}$ be a continuous function, and suppose that the set $K = \{(X^{(1)}(\mathbf{w}), X^{(2)}(\mathbf{w}), \dots, X^{(k)}(\mathbf{w})) : \mathbf{w} \in \text{supp}(\mathbf{w})\}$ is compact in $\mathbb{R}^k$. Assuming that,*

$$\max_{1 \leq i \leq k} \sup_{\mathbf{w} \in \text{supp}(\mathbf{w})} |X_n^{(i)}(\mathbf{w}) - X^{(i)}(\mathbf{w})| \xrightarrow{p} 0 \quad \text{as } n \rightarrow \infty.$$

Then, as $n \rightarrow \infty$,

$$\sup_{\mathbf{w} \in \text{supp}(\mathbf{w})} |g(X_n^{(1)}(\mathbf{w}), \dots, X_n^{(k)}(\mathbf{w})) - g(X^{(1)}(\mathbf{w}), \dots, X^{(k)}(\mathbf{w}))| \xrightarrow{p} 0.$$

C.5 Validity of Bootstrap Inference

The following theorems establish the consistency of the bootstrap distribution in estimating the sampling distribution of bagged nearest neighbors estimators, by showing that both distributions converge uniformly to the same normal limit. This result validates the use of bootstrap variance for confidence interval construction.

Theorem C.8 (Uniform Bootstrap CLT). *Let $\{(\mathbf{w}_i, y_i)\}_{i=1}^n$ be an i.i.d. sample from the distribution of $(\mathbf{w}, y)$, and let $\mathbb{P}_n$ denote the corresponding empirical distribution. Define $\{(\mathbf{w}_i^*, y_i^*)\}_{i=1}^n$ as the bootstrap sample drawn independently from $\mathbb{P}_n$. Let the bootstrap version of the bagged nearest neighbors estimator be*

$$\tau_n^*(m)(\mathbf{w}) = \binom{n}{m}^{-1} \sum_{1 \leq i_1 < \dots < i_m \leq n} \Phi(\mathbf{w}; Z_{i_1}^*, \dots, Z_{i_m}^*),$$

where $Z_i^* = (\mathbf{w}_i^*, y_i^*)$ and $\Phi(\mathbf{w}; Z_{i_1}^*, \dots, Z_{i_m}^*)$ denotes the corresponding response of the near-

est neighbor to $\mathbf{w}$ in $\{Z_{i_1}^*, \dots, Z_{i_m}^*\}$. Suppose Assumptions C.1–C.5 hold and let $\sup_{\mathbf{w}} \sigma_n^2(\mathbf{w}) = O(m/n)$ and $\inf_{\mathbf{w}} \sigma^2(\mathbf{w}) \geq c > 0$ be as defined in Theorem C.2. Then, as $n \rightarrow \infty$, $m \rightarrow \infty$, and $m/n \rightarrow 0$, the bootstrap estimator satisfies uniform asymptotic normality:

$$\sup_{\mathbf{w} \in \text{supp}(\mathbf{w})} \sup_{x \in \mathbb{R}} \left| \mathbb{P}_n^* \left(\frac{\tau_n^*(m)(\mathbf{w}) - \mathbb{E}_{\mathbb{P}_n^*}[\tau_n^*(m)(\mathbf{w})]}{\sigma_n^*(\mathbf{w})} \leq x \right) - \Phi(x) \right| \xrightarrow{p} 0,$$

where $\mathbb{P}_n^*$ and $\mathbb{E}_{\mathbb{P}_n^*}$ denote probability and expectation with respect to the bootstrap distribution, conditional on the original sample, and $\Phi(x)$ is the standard normal cumulative distribution function.

Theorem C.9 (Uniform Bootstrap Validity for Bagged Nearest Neighbors). *Let $(\mathbf{w}_i, y_i)_{i=1}^n$ be an i.i.d. sample from the distribution of $(\mathbf{w}, y)$, and let $\mathbb{P}_n$ denote the corresponding empirical distribution. Define $(\mathbf{w}_i^*, y_i^*)_{i=1}^n$ as the bootstrap sample drawn independently from $\mathbb{P}_n$. Let the bootstrap version of the bagged nearest neighbors estimator be:*

$$\tau_n^*(m)(\mathbf{w}) = \binom{n}{m}^{-1} \sum_{1 \leq i_1 < \dots < i_m \leq n} \Phi(\mathbf{w}; Z_{i_1}^*, \dots, Z_{i_m}^*),$$

where $Z_i^* = (\mathbf{w}_i^*, y_i^*)$ and $\Phi(\mathbf{w}; Z_{i_1}^*, \dots, Z_{i_m}^*)$ denotes the corresponding response of the nearest neighbor to $\mathbf{w}$ in $Z_{i_1}^*, \dots, Z_{i_m}^*$. Suppose Assumptions C.1–C.5 hold. Let $\sup_{\mathbf{w}} \sigma_n^2(\mathbf{w}) = O(m/n)$ and $\inf_{\mathbf{w}} \sigma^2(\mathbf{w}) \geq c > 0$ be as defined in Theorem C.2. Then, as $n \rightarrow \infty$, $m \rightarrow \infty$, and $m/n \rightarrow 0$, the bootstrap consistently estimates the sampling distribution of the normalized estimator uniformly over $\mathbf{w} \in \text{supp}(\mathbf{w})$.

$$\sup_{\mathbf{w} \in \text{supp}(\mathbf{w})} \sup_{x \in \mathbb{R}} |F_n^*(x, \mathbf{w}) - F_n(x, \mathbf{w})| \xrightarrow{p} 0,$$

where $F_n^*(x, \mathbf{w})$ and $F_n(x, \mathbf{w})$ are the CDFs of the normalized bootstrap and original estimators:

$$F_n^*(x, \mathbf{w}) = \mathbb{P}_n^* \left(\frac{\tau_n^*(m)(\mathbf{w}) - \mathbb{E}_{\mathbb{P}_n^*}[\tau_n^*(m)(\mathbf{w})]}{\sigma_n^*(\mathbf{w})} \leq x \right),$$

$$F_n(x, \mathbf{w}) = \mathbb{P} \left(\frac{\tau_n(m)(\mathbf{w}) - \mathbb{E}[\tau_n(m)(\mathbf{w})]}{\sigma_n(\mathbf{w})} \leq x \right).$$

Here, $\mathbb{P}_n^*$ and $\mathbb{E}_{\mathbb{P}_n^*}$ denote probability and expectation with respect to the bootstrap distribution, conditional on the original sample.

Lemma 2 (Bootstrap Continuous Mapping Theorem). *Let $\{\hat{\theta}_n(\mathbf{w})\}$ be a sequence of random processes indexed by $\mathbf{w} \in \text{supp}(\mathbf{w})$ taking values in $\mathbb{R}^k$, and let $\{\hat{\theta}_n^*(\mathbf{w})\}$ be its bootstrap*

version. Let $r_n \rightarrow \infty$ be a sequence of constants (the convergence rate). Let $g : \mathbb{R}^k \rightarrow \mathbb{R}$ be a function that is Hadamard differentiable at the true parameter $\theta(\cdot)$ uniformly over $\text{supp}(\mathbf{w})$, with derivative $g'_{\theta(\cdot)}$. Suppose that:

1. **Weak Convergence of the Process:** The scaled empirical process converges weakly to a tight Gaussian process $\mathbb{G}$:

$$r_n(\hat{\theta}_n(\cdot) - \theta(\cdot)) \rightsquigarrow \mathbb{G}(\cdot) \quad \text{in } \ell^\infty(\text{supp}(\mathbf{w})).$$

2. **Bootstrap Validity of the Process:** Conditional on the original sample, the bootstrap process converges to the same limit:

$$r_n(\hat{\theta}_n^*(\cdot) - \hat{\theta}_n(\cdot)) \rightsquigarrow \mathbb{G}(\cdot) \quad \text{in probability.}$$

Then, the bootstrap distribution of the transformed estimator $g(\hat{\theta}_n)$ consistently estimates the limit distribution of $g(\hat{\theta}_n)$:

$$\sup_{x \in \mathbb{R}} \left| \mathbb{P}_n^* \left(r_n \frac{g(\hat{\theta}_n^*(\mathbf{w})) - g(\hat{\theta}_n(\mathbf{w}))}{\sigma_g(\mathbf{w})} \leq x \right) - \Phi(x) \right| \xrightarrow{p} 0 \quad \text{uniformly in } \mathbf{w},$$

where $\sigma_g(\mathbf{w})$ is the standard deviation of the limit process $g'_{\theta(\mathbf{w})}(\mathbb{G}(\mathbf{w}))$.

Theorem C.10 (Bootstrap Validity for Finite Differences of BNN). Let $\tau_n(m)(\mathbf{w})$ be a bagged nearest neighbors estimator satisfying Assumptions C.1–C.5. Define the finite difference estimator:

$$\Delta_{h_n} \tau_n(m)(\mathbf{w}) = \frac{\tau_n(m)(\mathbf{w} + h_n \mathbf{e}_j) - \tau_n(m)(\mathbf{w})}{h_n},$$

and its bootstrap version:

$$\Delta_{h_n} \tau_n^*(m)(\mathbf{w}) = \frac{\tau_n^*(m)(\mathbf{w} + h_n \mathbf{e}_j) - \tau_n^*(m)(\mathbf{w})}{h_n}.$$

Assume that as $n \rightarrow \infty$, $m \rightarrow \infty$, and $m/n \rightarrow 0$, the bandwidth sequence satisfies $h_n \rightarrow 0$ and $m^{4/d} h_n \rightarrow \infty$. Then:

$$\sup_{\mathbf{w} \in \text{supp}(\mathbf{w})} \sup_{x \in \mathbb{R}} |\mathbb{P}_n^*(\Delta_{h_n} \tau_n^*(m)(\mathbf{w}) \leq x) - \mathbb{P}(\Delta_{h_n} \tau_n(m)(\mathbf{w}) \leq x)| \xrightarrow{p} 0,$$

where $\mathbb{P}_n^*$ denotes the bootstrap probability conditional on the original sample.

Theorem C.11 (Uniform Bootstrap Validity for Transformations). *Consider two types of estimators:*

1. *A collection of k_1 bagged nearest neighbors estimators $\{\tau_{n,i}(m_i)(\mathbf{w})\}_{i=1}^{k_1}$,*
2. *A collection of k_2 derivative estimators $\{\Delta\tau_{n,j}(m_j)(\mathbf{w})\}_{j=1}^{k_2}$.*

Let $g : \mathbb{R}^{k_1+k_2} \rightarrow \mathbb{R}$ be Hadamard differentiable at the true parameter vector $\theta(\mathbf{w})$ uniformly in $\mathbf{w} \in \text{supp}(\mathbf{w})$ with non-zero derivative. Define the transformed estimator:

$$T_n(\mathbf{w}) = g(\tau_{n,1}(m_1)(\mathbf{w}), \dots, \Delta\tau_{n,k_2}(m_{k_2})(\mathbf{w})).$$

Let $T_n^(\mathbf{w})$ be the bootstrap version constructed by replacing each estimator with its bootstrap counterpart. Let $\hat{\sigma}_n^*(\mathbf{w})$ be the bootstrap estimate of the standard deviation of $T_n^*(\mathbf{w})$. Assume:*

1. *Each BNN estimator satisfies Assumptions C.1–C.5 with subsample size $m_i \rightarrow \infty$ and $m_i/n \rightarrow 0$,*
2. *The finite difference step size h_n satisfies $h_n \rightarrow 0$ and $m_i^{4/d}h_n \rightarrow \infty$.*

Then the bootstrap is uniformly valid for the studentized transformed estimator:

$$\sup_{\mathbf{w} \in \text{supp}(\mathbf{w})} \sup_{x \in \mathbb{R}} \left| \mathbb{P}_n^* \left(\frac{T_n^*(\mathbf{w}) - T_n(\mathbf{w})}{\hat{\sigma}_n^*(\mathbf{w})} \leq x \right) - \Phi(x) \right| \xrightarrow{p} 0,$$

where $\mathbb{P}_n^$ denotes probability under the bootstrap distribution conditional on the original sample, and $\Phi(x)$ is the standard normal CDF.*

C.6 Proof of theorems

C.6.1 Proof of Theorem C.1 and C.2

For a complete proof, readers are referred to Theorem 1 and 2 in Demirkaya et al. (2022).

C.6.2 Proof of Theorem C.3 using Concentration Inequality

In this subsection, we provide two proofs for Theorem C.3, which establishes the uniform convergence of the bagged nearest neighbors estimator.

Lemma 3 (Concentration Inequality for U-statistics). *Let U_n be a U-statistic of order m with bounded kernel $|\Phi| \leq B$. If $n \rightarrow \infty$, $m \rightarrow \infty$, and $m/n \rightarrow 0$, then for any $t > 0$,*

$$\mathbb{P}(|U_n - \mathbb{E}[U_n]| > t) \leq 2 \exp \left(-\frac{t^2}{2B^2} \frac{n}{m} \right).$$

This lemma provides an improvement over the Hoeffding's concentration inequality for U-statistics, while Hoeffding's original bound for a U-statistic U_n of order m with bounded kernel $|\Phi| \leq B$ gives:

$$\mathbb{P}(|U_n - \mathbb{E}[U_n]| > t) \leq 2 \exp \left(-\frac{t^2}{2B^2} \frac{n}{m^2} \right).$$

This refinement is particularly valuable when analyzing U-statistics with diverging order, where m increases with n . The improved bound provides tighter control over the concentration of U-statistics in such scenarios. If we use Hoeffding's classical concentration bound for U-statistics instead of the sharper bound provided in Lemma 3, we would require the stronger condition $m^2/n \rightarrow 0$ to show uniform convergence. We next present an elegant proof of this lemma, adapted from Professor Alessandro Rinaldo's Fall 2016 Advanced Statistical Theory lecture notes.¹ We do not contribute to this result.

Proof. Let $k = \lceil \frac{n}{m} \rceil$, the smallest integer $\geq \frac{n}{m}$, and define

$$W(x_1, \dots, x_n) = \frac{\Phi(x_1, \dots, x_m) + \Phi(x_{m+1}, \dots, x_{2m}) + \dots + \Phi(x_{m(k-1)+1}, \dots, x_{mk})}{k},$$

¹https://www.stat.cmu.edu/~arinaldo/Teaching/36755/F16/Scribed_Lectures/755_dec5.pdf (last accessed July 5, 2024).

which breaks our sample into k non-overlapping blocks of size m . Then

$$k \sum_{\sigma \in S_n} W(X_{\sigma_1}, \dots, X_{\sigma_n}) = km!(n-m)! \sum_{1 \leq i_1 < \dots < i_m \leq n} \Phi(X_{i_1}, \dots, X_{i_m}),$$

where S_n is the permutation group on $1, \dots, n$. This gives us

$$\sum_{\sigma \in S_n} W(X_{\sigma_1}, \dots, X_{\sigma_n}) = m!(n-m)! \binom{n}{m} U_n = n! U_n,$$

which implies

$$U_n = \frac{1}{n!} \sum_{\sigma \in S_n} W(X_{\sigma_1}, \dots, X_{\sigma_n}).$$

For a fixed σ , $W(X_{\sigma_1}, \dots, X_{\sigma_n})$ is an average of k i.i.d. random variables bounded by B in absolute value. Define

$$T_\sigma = W(X_{\sigma_1}, \dots, X_{\sigma_n}) - \mathbb{E}[U_n] \in \text{SubGaussian} \left(\frac{B^2}{k} \right).$$

Now, we have

$$U_n - \mathbb{E}[U_n] = \frac{1}{n!} \sum_{\sigma} T_\sigma.$$

For $t > 0$, using Markov's inequality and Jensen's inequality:

$$\begin{aligned} \mathbb{P}(U_n - \mathbb{E}[U_n] \geq t) &\leq e^{-\lambda t} \mathbb{E}[e^{\lambda(U_n - \mathbb{E}[U_n])}] \quad \text{for any } \lambda > 0 \\ &= e^{-\lambda t} \mathbb{E} \left[e^{\lambda \frac{1}{n!} \sum_{\sigma} T_\sigma} \right] \\ &\leq e^{-\lambda t} \frac{1}{n!} \sum_{\sigma} \mathbb{E}[e^{\lambda T_\sigma}] \\ &\leq e^{-\lambda t} e^{\frac{\lambda^2 B^2}{2k}} \quad \text{since } T_\sigma \in \text{SubGaussian} \left(\frac{B^2}{k} \right) \\ &= e^{-\frac{t^2}{2B^2} k} \quad \text{taking } \lambda = \frac{kt}{B^2} \text{ (minimizing over } \lambda) \\ &\leq \exp \left(-\frac{t^2}{2B^2} \frac{n}{m} \right). \end{aligned}$$

Repeating the argument for the other side of the inequality and combining the results gives

$$\mathbb{P}(|U_n - \mathbb{E}[U_n]| > t) \leq 2 \exp \left(-\frac{t^2}{2B^2} \frac{n}{m} \right),$$

which completes the proof. $\square$

Proof of Theorem C.3. In this proof, we utilize the high probability concentration bound for U-statistics. Given $\mathbf{w}_0 \in \text{supp}(\mathbf{w})$, under Assumptions C.1–C.5, we proceed as follows:

Step 1: Decomposition

$$|\tau_n(m)(\mathbf{w}_0) - \mu(\mathbf{w}_0)| \leq |\tau_n(m)(\mathbf{w}_0) - \mathbb{E}[\tau_n(m)(\mathbf{w}_0)]| + |\mathbb{E}[\tau_n(m)(\mathbf{w}_0)] - \mu(\mathbf{w}_0)|.$$

Step 2: Bias Analysis From Theorem C.1, $B(m) = O(m^{-2/d})$ uniformly in $\mathbf{w}_0$ due to the compactness of $\text{supp}(\mathbf{w})$ and boundedness of derivatives. Thus for every $\epsilon > 0$ as $m \rightarrow \infty$,

$$A_n := \mathbb{P} \left(\sup_{\mathbf{w}_0 \in \text{supp}(\mathbf{w})} |\mathbb{E}[\tau_n(m)(\mathbf{w}_0)] - \mu(\mathbf{w}_0)| > \frac{\epsilon}{2} \right) \rightarrow 0.$$

Step 3: Sharp concentration bound By Lemma 3, for any fixed $\mathbf{w}_0$, we have the exponential bound $2 \exp(-nt^2/(2mB^2))$. Since $\text{supp}(\mathbf{w})$ is compact and the kernel is bounded, we cover the support with a finite grid $\mathcal{W}_n$ of size K_n . We choose the grid fine enough such that the discretization error is at most $\epsilon/4$. By the union bound:

$$\begin{aligned} B_n &:= \mathbb{P} \left(\sup_{\mathbf{w}_0 \in \text{supp}(\mathbf{w})} |\tau_n(m)(\mathbf{w}_0) - \mathbb{E}[\tau_n(m)(\mathbf{w}_0)]| > \frac{\epsilon}{2} \right) \\ &\leq \sum_{\mathbf{w} \in \mathcal{W}_n} \mathbb{P} \left(|\tau_n(m)(\mathbf{w}) - \mathbb{E}[\tau_n(m)(\mathbf{w})]| > \frac{\epsilon}{4} \right) \\ &\leq K_n \cdot 2 \exp \left(-\frac{n(\epsilon/4)^2}{2mB^2} \right) = 2K_n \exp \left(-\frac{n\epsilon^2}{32mB^2} \right). \end{aligned}$$

Since the covering number K_n grows polynomially while the probability decays exponentially (as $m/n \rightarrow 0$), the upper bound converges to 0.

Step 4: Combining results By the union bound and triangular inequality, as $n \rightarrow \infty$, $m \rightarrow \infty$, $m/n \rightarrow 0$, for every $\epsilon > 0$,

$$\mathbb{P} \left(\sup_{\mathbf{w}_0 \in \text{supp}(\mathbf{w})} |\tau_n(m)(\mathbf{w}_0) - \mu(\mathbf{w}_0)| > \epsilon \right) \leq A_n + B_n \rightarrow 0.$$

This proves uniform convergence in probability of $\tau_n(m)(\mathbf{w}_0)$ to $\mu(\mathbf{w}_0)$ over $\text{supp}(\mathbf{w})$. $\square$

The uniform convergence result presented above can also be established using the stochastic equicontinuity and Lipschitz conditions framework introduced by Newey (1991). However, the above proof exploits U-statistic concentration inequalities, resulting in a simplified proof that leverages on the specific structure of the bagged nearest neighbors estimator. The uniform bounds serve a similar purpose to stochastic equicontinuity by controlling the estimator's fluctuations uniformly over its support. For completeness, we also present a formal proof following Newey (1991), which brings additional insights into the result.

C.6.3 Proof of Theorem C.3 using Stochastic Equicontinuity

Proof. We formally show another proof of Theorem C.3 using the structure of Theorem 2.1 in Newey (1991). In this proof, we show that our estimator is stochastic equicontinuous.

Compactness: By Assumption C.2, the support of $\mathbf{w}$, denoted as $\text{supp}(\mathbf{w})$, is compact.

Point-wise Convergence: From Theorem C.1 and Theorem C.2, we have pointwise convergence in probability. Specifically, for any fixed $\mathbf{w}_0 \in \text{supp}(\mathbf{w})$:

$$\tau_n(m)(\mathbf{w}_0) - \mu(\mathbf{w}_0) = B(m) + O_p(\sigma_n) = o_p(1).$$

This is because $B(m) = O(m^{-2/d}) = o(1)$ as $m \rightarrow \infty$, and $\sigma_n = O(\sqrt{m/n}) = o(1)$ as $m/n \rightarrow 0$. This satisfies Assumption 2 (point-wise convergence) in Newey's paper.

Stochastic Equicontinuity: We need to show that for every $\varepsilon, \eta > 0$, there exists a random variable $\Delta_n(\varepsilon, \eta)$ and a constant $n_0(\varepsilon, \eta)$ such that for $n > n_0(\varepsilon, \eta)$: a) $P(\Delta_n(\varepsilon, \eta) > \varepsilon) < \eta$ and b) for each $\mathbf{w}_0$, there is an open set $N(\mathbf{w}_0, \varepsilon, \eta)$ containing $\mathbf{w}_0$ with:

$$\sup_{\mathbf{w} \in N(\mathbf{w}_0, \varepsilon, \eta)} |\tau_n(m)(\mathbf{w}) - \tau_n(m)(\mathbf{w}_0)| < \Delta_n(\varepsilon, \eta).$$

By Assumption C.3, $\mu(\cdot)$ is twice continuously differentiable with bounded second derivatives on the compact set $\text{supp}(\mathbf{w})$. This implies $\mu(\cdot)$ is Lipschitz continuous on $\text{supp}(\mathbf{w})$. Let L be the Lipschitz constant for $\mu(\cdot)$. For any $\mathbf{w}, \mathbf{w}_0 \in \text{supp}(\mathbf{w})$, we have:

$$\begin{aligned} |\tau_n(m)(\mathbf{w}) - \tau_n(m)(\mathbf{w}_0)| &\leq |\tau_n(m)(\mathbf{w}) - \mu(\mathbf{w})| + |\mu(\mathbf{w}) - \mu(\mathbf{w}_0)| + |\mu(\mathbf{w}_0) - \tau_n(m)(\mathbf{w}_0)| \\ &\leq |\tau_n(m)(\mathbf{w}) - \mu(\mathbf{w})| + L\|\mathbf{w} - \mathbf{w}_0\| + |\mu(\mathbf{w}_0) - \tau_n(m)(\mathbf{w}_0)|. \end{aligned}$$

Let $B_n = \sup_{\mathbf{w} \in \text{supp}(\mathbf{w})} |\tau_n(m)(\mathbf{w}) - \mu(\mathbf{w})|$. Then:

$$|\tau_n(m)(\mathbf{w}) - \tau_n(m)(\mathbf{w}_0)| \leq 2B_n + L\|\mathbf{w} - \mathbf{w}_0\|.$$

From Theorem C.1 and C.2, we have pointwise convergence. Since $\text{supp}(\mathbf{w})$ is compact, for any ε , we can cover it with a finite grid of points. The maximum error over this finite grid is $o_p(1)$. Combined with the Lipschitz continuity of the true function $\mu(\cdot)$, this controls the oscillations between grid points. Consequently, the uniform error B_n is dominated by the error on this grid plus a negligible oscillation term. Thus, $B_n = O_p(m^{-2/d} + \sqrt{m/n}) = o_p(1)$ under Assumption C.5. This establishes an asymptotic Lipschitz condition that implies stochastic equicontinuity. Specifically, define $\Delta_n(\varepsilon, \eta) = 2B_n + L\delta$, where $\delta = \varepsilon/(2L)$. Since $B_n = o_p(1)$, for any $\varepsilon > 0$ and $\eta > 0$, we can find $n_0(\varepsilon, \eta)$ such that for all $n > n_0(\varepsilon, \eta)$, $P(2B_n > \varepsilon/2) < \eta/2$. Then condition (a) of stochastic equicontinuity is satisfied:

$$P(\Delta_n(\varepsilon, \eta) > \varepsilon) = P(2B_n + L\delta > \varepsilon) = P(2B_n > \varepsilon/2) < \eta/2 < \eta.$$

For condition (b), define $N(\mathbf{w}_0, \varepsilon, \eta) = \{\mathbf{w} : \|\mathbf{w} - \mathbf{w}_0\| < \delta\}$. For any $\mathbf{w} \in N(\mathbf{w}_0, \varepsilon, \eta)$:

$$\sup_{\mathbf{w} \in N(\mathbf{w}_0, \varepsilon, \eta)} |\tau_n(m)(\mathbf{w}) - \tau_n(m)(\mathbf{w}_0)| \leq 2B_n + L\|\mathbf{w} - \mathbf{w}_0\| < 2B_n + L\delta = \Delta_n(\varepsilon, \eta).$$

This establishes Assumption 3 (stochastic equicontinuity) in Newey's paper.

Equicontinuity of $\mu(\mathbf{w})$: This follows directly from the Lipschitz continuity of $\mu(\cdot)$.

Apply Theorem 2.1 from Newey (1991): Given that all conditions of Theorem 2.1 are satisfied, we can conclude:

$$\sup_{\mathbf{w}_0 \in \text{supp}(\mathbf{w})} |\tau_n(m)(\mathbf{w}_0) - \mu(\mathbf{w}_0)| \xrightarrow{p} 0.$$

This establishes the uniform convergence in probability stated in our theorem. Therefore, under Assumptions C.1–C.5, we have proven that for every $\epsilon > 0$,

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(\sup_{\mathbf{w}_0 \in \text{supp}(\mathbf{w})} |\tau_n(m)(\mathbf{w}_0) - \mu(\mathbf{w}_0)| > \epsilon \right) = 0,$$

where $m \rightarrow \infty$ and $m/n \rightarrow 0$ as $n \rightarrow \infty$. □

C.6.4 Proof of Theorem C.4 using the uniform CLT

Proof of Theorem C.4. We proceed in several steps:

Step 1: Hoeffding Decomposition First, we write the centered and scaled estimator in terms of its Hoeffding decomposition:

$$\frac{\tau_n(m)(\mathbf{w}) - \mu(\mathbf{w}) - B(m)}{\sigma_n(\mathbf{w})} = \frac{\tau_n(m)(\mathbf{w}) - \mathbb{E}[\tau_n(m)(\mathbf{w})]}{\sigma_n(\mathbf{w})} + \frac{\mathbb{E}[\tau_n(m)(\mathbf{w})] - \mu(\mathbf{w}) - B(m)}{\sigma_n(\mathbf{w})}.$$

The bagged nearest neighbors estimator can be written as a U-statistic:

$$\tau_n(m)(\mathbf{w}) = \binom{n}{m}^{-1} \sum_{1 \leq i_1 < \dots < i_m \leq n} \Phi(\mathbf{w}; Z_{i_1}, \dots, Z_{i_m}).$$

By Hoeffding's decomposition:

$$\tau_n(m)(\mathbf{w}) - \mathbb{E}[\tau_n(m)(\mathbf{w})] = \frac{m}{n} \sum_{i=1}^n g_1(Z_i; \mathbf{w}) + \frac{m(m-1)}{n(n-1)} \sum_{1 \leq i < j \leq n} g_2(Z_i, Z_j; \mathbf{w}) + R_n(\mathbf{w}),$$

where $g_1(z; \mathbf{w}) = \mathbb{E}[\Phi(\mathbf{w}; Z_1, \dots, Z_m) | Z_1 = z] - \mathbb{E}[\Phi(\mathbf{w}; Z_1, \dots, Z_m)]$ is the first-order term, $g_2(z_1, z_2; \mathbf{w})$ is the second-order term, and $R_n(\mathbf{w})$ contains the higher-order terms.

Step 2: Analysis of the Leading Term Consider the leading term $(m/n) \sum_{i=1}^n g_1(Z_i; \mathbf{w})$. Let $\mathcal{F} = \{g_1(\cdot; \mathbf{w}) : \mathbf{w} \in \text{supp}(\mathbf{w})\}$. We verify the conditions of the bracketing uniform CLT:

1. By Assumptions C.3 and C.4, $g_1(z; \mathbf{w})$ is bounded and continuously differentiable in its arguments with bounded derivatives on the compact $\text{supp}(\mathbf{w})$. This implies that $g_1(z; \mathbf{w})$ is Lipschitz continuous with a uniform Lipschitz constant. Since $\mathcal{F}$ is a class of bounded Lipschitz functions on a compact index set in $\mathbb{R}^d$, it has a bracketing number that satisfies $N_{[]}(\epsilon, \mathcal{F}, L_2(P)) \leq C\epsilon^{-d}$ for some constant $C > 0$. Therefore, the bracketing entropy integral condition is satisfied:

$$\int_0^\delta \sqrt{\log N_{[]}(\epsilon, \mathcal{F}, L_2(P))} d\epsilon < \infty$$

for any $\delta > 0$, as this integral converges for any (fixed) finite dimension d .

2. $\mathbb{E}[g_1(Z; \mathbf{w})] = 0$ for all $\mathbf{w} \in \text{supp}(\mathbf{w})$ by construction.
3. $\sup_{\mathbf{w} \in \text{supp}(\mathbf{w})} \mathbb{E}[g_1^2(Z; \mathbf{w})] < \infty$ by Assumption C.4.

Therefore, by the bracketing central limit theorem for empirical processes, the normalized empirical process converges weakly to a Gaussian process, which implies:

$$\sup_{\mathbf{w} \in \text{supp}(\mathbf{w})} \sup_{x \in \mathbb{R}} \left| \mathbb{P} \left(\frac{m}{n\sigma_n(\mathbf{w})} \sum_{i=1}^n g_1(Z_i; \mathbf{w}) \leq x \right) - \Phi(x) \right| \rightarrow 0.$$

Step 3: Uniform Negligibility of Remainder Terms For the second-order term:

$$\sup_{\mathbf{w} \in \text{supp}(\mathbf{w})} \left| \frac{m(m-1)}{n(n-1)\sigma_n(\mathbf{w})} \sum_{1 \leq i < j \leq n} g_2(Z_i, Z_j; \mathbf{w}) \right| = O_p \left(\frac{m}{n\sigma_n(\mathbf{w})} \right) = o_p(1),$$

since $m/n \rightarrow 0$ and $\sup_{\mathbf{w}} \sigma_n^2(\mathbf{w}) = O(m/n)$ and $\inf_{\mathbf{w}} \sigma^2(\mathbf{w}) \geq c > 0$. For the remainder term:

$$\sup_{\mathbf{w} \in \text{supp}(\mathbf{w})} \left| \frac{R_n(\mathbf{w})}{\sigma_n(\mathbf{w})} \right| = O_p \left(\left(\frac{m}{n} \right)^{3/2} / \sigma_n(\mathbf{w}) \right) = o_p(1).$$

Step 4: Bias Analysis From Theorem C.1, we have:

$$\sup_{\mathbf{w} \in \text{supp}(\mathbf{w})} \left| \frac{\mathbb{E}[\tau_n(m)(\mathbf{w})] - \mu(\mathbf{w}) - B(m)}{\sigma_n(\mathbf{w})} \right| = o(1).$$

Step 5: Combining Results By Steps 3 and 4, all higher order terms and bias terms are uniformly $o_p(1)$, we have

$$\sup_{\mathbf{w} \in \text{supp}(\mathbf{w})} \sup_{x \in \mathbb{R}} \left| \mathbb{P} \left(\frac{\tau_n(m)(\mathbf{w}) - \mu(\mathbf{w}) - B(m)}{\sigma_n(\mathbf{w})} \leq x \right) - \Phi(x) \right| \rightarrow 0.$$

□

C.6.5 Proof of Theorem C.5

Proof. Let $\epsilon > 0$ be given. We aim to show that, as $n \rightarrow \infty$, $m \rightarrow \infty$, $m/n \rightarrow 0$,

$$\mathbb{P} \left(\sup_{\mathbf{w} \in \text{supp}(\mathbf{w})} \left| \Delta_{h_n} \tau_n(m)(\mathbf{w}) - \frac{\partial \mu}{\partial w_j}(\mathbf{w}) \right| > \epsilon \right) \rightarrow 0.$$

First, for any $\mathbf{w} \in \text{supp}(\mathbf{w})$, decompose the error as [Hong et al. \(2015\)](#):

$$\begin{aligned} \Delta_{h_n} \tau_n(m_n)(\mathbf{w}) - \frac{\partial \mu}{\partial w_j}(\mathbf{w}) &= \frac{\tau_n(m_n)(\mathbf{w} + h_n \mathbf{e}_j) - \mu(\mathbf{w} + h_n \mathbf{e}_j)}{h_n} (:= I_n(\mathbf{w})) \\ &\quad - \frac{\tau_n(m_n)(\mathbf{w}) - \mu(\mathbf{w})}{h_n} (:= II_n(\mathbf{w})) \\ &\quad + \frac{\mu(\mathbf{w} + h_n \mathbf{e}_j) - \mu(\mathbf{w})}{h_n} - \frac{\partial \mu}{\partial w_j}(\mathbf{w}) (:= III_n(\mathbf{w})). \end{aligned}$$

Then,

$$\left| \Delta_{h_n} \tau_n(m_n)(\mathbf{w}) - \frac{\partial \mu}{\partial w_j}(\mathbf{w}) \right| \leq |I_n(\mathbf{w}) - II_n(\mathbf{w})| + |III_n(\mathbf{w})|.$$

Bounding $I_n(\mathbf{w}) - II_n(\mathbf{w})$: From Theorem [C.3](#), we have

$$\begin{aligned} \tau_n(m)(\mathbf{w} + h_n \mathbf{e}_j) - \mu(\mathbf{w} + h_n \mathbf{e}_j) &= c(\mathbf{w} + h_n \mathbf{e}_j) m^{-2/d} + O_p(m^{-4/d}), \\ \tau_n(m)(\mathbf{w}) - \mu(\mathbf{w}) &= c(\mathbf{w}) m^{-2/d} + O_p(m^{-4/d}). \end{aligned}$$

$$I_n(\mathbf{w}) - II_n(\mathbf{w}) = \frac{c'(\mathbf{w}) h_n m^{-2/d} + c''(\xi) h_n^2 m^{-2/d} + O(m^{-4/d})}{h_n} = O(m^{-2/d}) + O_p\left(\frac{m^{-4/d}}{h_n}\right).$$

Bounding $III_n(\mathbf{w})$: Since $\mu(\mathbf{w})$ has continuous third-order partial derivatives on $\text{supp}(\mathbf{w})$, by Taylor's theorem, for each $\mathbf{w} \in \text{supp}(\mathbf{w})$,

$$\mu(\mathbf{w} + h_n \mathbf{e}_j) = \mu(\mathbf{w}) + h_n \frac{\partial \mu}{\partial w_j}(\mathbf{w}) + \frac{h_n^2}{2} \frac{\partial^2 \mu}{\partial w_j^2}(\mathbf{w}) + \frac{h_n^3}{6} \frac{\partial^3 \mu}{\partial w_j^3}(\xi_{\mathbf{w}}),$$

for some ξ between $\mathbf{w}$ and $\mathbf{w} + h_n \mathbf{e}_j$. Thus,

$$III_n(\mathbf{w}) = \frac{h_n}{2} \frac{\partial^2 \mu}{\partial w_j^2}(\mathbf{w}) + \frac{h_n^2}{6} \frac{\partial^3 \mu}{\partial w_j^3}(\xi) = O(h_n).$$

Since $\text{supp}(\mathbf{w})$ is compact, for sufficiently large n , h_n is small enough that $\mathbf{w} + h_n \mathbf{e}_j \in \text{supp}(\mathbf{w})$ for all $\mathbf{w} \in \text{supp}(\mathbf{w})$. We have:

$$\sup_{\mathbf{w} \in \text{supp}(\mathbf{w})} \left| \Delta_{h_n} \tau_n(m_n)(\mathbf{w}) - \frac{\partial \mu}{\partial w_j}(\mathbf{w}) \right| \leq O(m^{-2/d}) + O_p\left(\frac{m^{-4/d}}{h_n}\right) + O(h_n).$$

Choosing h_n Appropriately: We need to choose h_n such that: $\frac{m^{-4/d}}{h_n} \rightarrow 0$ and $h_n \rightarrow 0$. For instance, let us choose $h_n = m^{-2/d}$. Then $\frac{m^{-4/d}}{h_n} = m^{-2/d} \rightarrow 0$, as $m \rightarrow \infty$. For debiased BNN estimator, the bound is $O(m^{-4/d}) + O_p\left(\frac{m^{-6/d}}{h_n}\right) + O(h_n)$, in this case, we need to choose

h_n such that: $\frac{m^{-6/d}}{h_n} \rightarrow 0$, and $h_n \rightarrow 0$. For instance, let us choose $h_n = m^{-3/d}$. Then, $\frac{m^{-6/d}}{h_n} = m^{-3/d} \rightarrow 0$, as $m \rightarrow \infty$.

Conclusion: With this h_n , the finite difference estimator is uniformly consistent:

$$\sup_{\mathbf{w} \in \text{supp}(\mathbf{w})} \left| \Delta_{h_n} \tau_n(m_n)(\mathbf{w}) - \frac{\partial \mu}{\partial w_j}(\mathbf{w}) \right| \rightarrow 0.$$

□

C.6.6 Proof of Theorem C.6

Proof. For the variance term of $I_n(\mathbf{w}) - II_n(\mathbf{w})$: From Theorem C.1, we know that the bias of $\tau_n(m)(\mathbf{w})$ is uniformly $O(m^{-2/d})$ and its variance is uniformly $O(m/n)$ over $\text{supp}(\mathbf{w})$.

$$\sup_{\mathbf{w} \in \text{supp}(\mathbf{w})} \text{Var}(I_n(\mathbf{w}) - II_n(\mathbf{w})) = O\left(\frac{m}{nh_n^2}\right).$$

For the bias term of $III_n(\mathbf{w})$: Using Taylor expansion and the continuity of third-order derivatives on the compact set $\text{supp}(\mathbf{w})$:

$$\sup_{\mathbf{w} \in \text{supp}(\mathbf{w})} |III_n(\mathbf{w})| = \sup_{\mathbf{w} \in \text{supp}(\mathbf{w})} \left| \frac{h_n}{2} \frac{\partial^2 \mu}{\partial w_j^2}(\mathbf{w}) + O(h_n^2) \right| = O(h_n).$$

Now, the uniform MSE is:

$$\sup_{\mathbf{w} \in \text{supp}(\mathbf{w})} \mathbb{E} \left[\left(\Delta_h \tau_n(m)(\mathbf{w}) - \frac{\partial \mu(\mathbf{w})}{\partial w_j} \right)^2 \right] = O\left(\frac{m}{nh_n^2}\right) + O(m^{-4/d}) + O\left(\frac{m^{-8/d}}{h_n^2}\right) + O(h_n^2).$$

To minimize this, we set the three terms equal: $\frac{m}{nh_n^2} = \frac{m^{-8/d}}{h_n^2} = h_n^2$. Solving these equations,

$$h_n = O(n^{-\frac{2}{d+8}}); \quad m = O(n^{\frac{d}{d+8}}).$$

If we use debiased BNN estimator to construct finite difference, the bias of debiased BNN estimator is uniformly $O(m^{-4/d})$ with variance also uniformly $O(m/n)$. In this case, we need these three terms to be equal: $\frac{m}{nh_n^2} = \frac{m^{-12/d}}{h_n^2} = h_n^2$. Solving these equations, we get:

$$h = O(n^{-\frac{3}{d+12}}); \quad m = O(n^{\frac{d}{d+12}}).$$

We verify that these choices satisfy the conditions in Theorem C.5. □

C.6.7 Proof of Theorem C.7

Proof. We proceed in several steps:

Step 1: From Theorem C.3, we have for $i = 1, 2$:

$$\sup_{\mathbf{w} \in \text{supp}(\mathbf{w})} |\tau_{n_i}^{(i)}(m_i)(\mathbf{w}) - \mu_i(\mathbf{w})| \xrightarrow{p} 0 \text{ as } n_i \rightarrow \infty, m_i \rightarrow \infty, m_i/n_i \rightarrow 0,$$

Step 2: We can write:

$$\begin{aligned} \hat{\theta}_n(\mathbf{w}) - \theta(\mathbf{w}) &= \frac{\tau_{n_1}^{(1)}(m_1)(\mathbf{w})}{\tau_{n_2}^{(2)}(m_2)(\mathbf{w})} - \frac{\mu_1(\mathbf{w})}{\mu_2(\mathbf{w})} \\ &= \frac{\tau_{n_1}^{(1)}(m_1)(\mathbf{w})\mu_2(\mathbf{w}) - \tau_{n_2}^{(2)}(m_2)(\mathbf{w})\mu_1(\mathbf{w})}{\tau_{n_2}^{(2)}(m_2)(\mathbf{w})\mu_2(\mathbf{w})} \\ &= \frac{\tau_{n_1}^{(1)}(m_1)(\mathbf{w})\mu_2(\mathbf{w}) - \mu_1(\mathbf{w})\mu_2(\mathbf{w}) + \mu_1(\mathbf{w})\mu_2(\mathbf{w}) - \tau_{n_2}^{(2)}(m_2)(\mathbf{w})\mu_1(\mathbf{w})}{\tau_{n_2}^{(2)}(m_2)(\mathbf{w})\mu_2(\mathbf{w})} \\ &= \frac{\mu_2(\mathbf{w})(\tau_{n_1}^{(1)}(m_1)(\mathbf{w}) - \mu_1(\mathbf{w})) + \mu_1(\mathbf{w})(\mu_2(\mathbf{w}) - \tau_{n_2}^{(2)}(m_2)(\mathbf{w}))}{\tau_{n_2}^{(2)}(m_2)(\mathbf{w})\mu_2(\mathbf{w})}. \end{aligned}$$

Step 3: Bounding the Denominator By the assumption that $|\mu_2(\mathbf{w})| > c > 0$ for all $\mathbf{w} \in \text{supp}(\mathbf{w})$ and the uniform convergence of $\tau_{n_2}^{(2)}(m_2)(\mathbf{w})$, we have that for any $\epsilon > 0$, there exists N_ϵ such that for all $n_2 > N_\epsilon$:

$$\mathbb{P} \left(\inf_{\mathbf{w} \in \text{supp}(\mathbf{w})} |\tau_{n_2}^{(2)}(m_2)(\mathbf{w})| > \frac{c}{2} \right) > 1 - \epsilon.$$

Step 4: Bounding the Numerator Let $M = \sup_{\mathbf{w} \in \text{supp}(\mathbf{w})} \max(|\mu_1(\mathbf{w})|, |\mu_2(\mathbf{w})|)$, which is finite due to the compactness of $\text{supp}(\mathbf{w})$ and continuity of μ_1 and μ_2 . Then:

$$\begin{aligned} &|\mu_2(\mathbf{w})(\tau_{n_1}^{(1)}(m_1)(\mathbf{w}) - \mu_1(\mathbf{w})) + \mu_1(\mathbf{w})(\mu_2(\mathbf{w}) - \tau_{n_2}^{(2)}(m_2)(\mathbf{w}))| \\ &\leq M|\tau_{n_1}^{(1)}(m_1)(\mathbf{w}) - \mu_1(\mathbf{w})| + M|\tau_{n_2}^{(2)}(m_2)(\mathbf{w}) - \mu_2(\mathbf{w})|. \end{aligned}$$

Step 5: Combining the Bounds For any $\delta > 0$, we have:

$$\begin{aligned}
& \mathbb{P} \left(\sup_{\mathbf{w} \in \text{supp}(\mathbf{w})} |\hat{\theta}_n(\mathbf{w}) - \theta(\mathbf{w})| > \delta \right) \\
& \leq \mathbb{P} \left(\sup_{\mathbf{w} \in \text{supp}(\mathbf{w})} \frac{M|\tau_{n_1}^{(1)}(m_1)(\mathbf{w}) - \mu_1(\mathbf{w})| + M|\tau_{n_2}^{(2)}(m_2)(\mathbf{w}) - \mu_2(\mathbf{w})|}{\frac{c}{2} \cdot c} > \delta \right) + \epsilon \\
& \leq \mathbb{P} \left(\sup_{\mathbf{w} \in \text{supp}(\mathbf{w})} |\tau_{n_1}^{(1)}(m_1)(\mathbf{w}) - \mu_1(\mathbf{w})| > \frac{c^2 \delta}{4M} \right) \\
& \quad + \mathbb{P} \left(\sup_{\mathbf{w} \in \text{supp}(\mathbf{w})} |\tau_{n_2}^{(2)}(m_2)(\mathbf{w}) - \mu_2(\mathbf{w})| > \frac{c^2 \delta}{4M} \right) + \epsilon.
\end{aligned}$$

Step 6: Conclusion As $\min(n_1, n_2) \rightarrow \infty$, $m_1, m_2 \rightarrow \infty$, and $m_1/n_1, m_2/n_2 \rightarrow 0$, each term in the last inequality converges to zero by the uniform convergence of the individual estimators and our choice of ϵ . Therefore:

$$\sup_{\mathbf{w} \in \text{supp}(\mathbf{w})} |\hat{\theta}_n(\mathbf{w}) - \theta(\mathbf{w})| \xrightarrow{p} 0.$$

This establishes the uniform consistency of the quotient estimator $\hat{\theta}_n(\mathbf{w})$. $\square$

C.6.8 Proof of Lemma 1 (Continuous Mapping Theorem)

Proof of Lemma 1. Let $\epsilon > 0$ be arbitrary. Since $g : \mathbb{R}^k \rightarrow \mathbb{R}$ is continuous and the set

$$K = \{ (X^{(1)}(\mathbf{w}), X^{(2)}(\mathbf{w}), \dots, X^{(k)}(\mathbf{w})) : \mathbf{w} \in \text{supp}(\mathbf{w}) \}$$

is compact by assumption, it follows that g is uniformly continuous on K . Therefore, there exists $\delta > 0$ such that for all $\mathbf{x}, \mathbf{y} \in K$,

$$\|\mathbf{x} - \mathbf{y}\|_1 = \sum_{i=1}^k |x_i - y_i| < \delta \implies |g(\mathbf{x}) - g(\mathbf{y})| < \epsilon.$$

Define the events, for each $i = 1, \dots, k$,

$$A_n^{(i)} = \left\{ \sup_{\mathbf{w} \in \text{supp}(\mathbf{w})} |X_n^{(i)}(\mathbf{w}) - X^{(i)}(\mathbf{w})| \geq \frac{\delta}{k} \right\}.$$

By the assumption of uniform convergence in probability,

$$P(A_n^{(i)}) \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Then, as $n \rightarrow \infty$, consider the event $A_n = \bigcup_{i=1}^k A_n^{(i)}$. Then,

$$P(A_n) \leq \sum_{i=1}^k P(A_n^{(i)}) \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

On the complement of A_n , i.e., when A_n^c occurs, we have for all $\mathbf{w}$ and $i = 1, \dots, k$,

$$|X_n^{(i)}(\mathbf{w}) - X^{(i)}(\mathbf{w})| < \frac{\delta}{k}.$$

Therefore,

$$\sum_{i=1}^k |X_n^{(i)}(\mathbf{w}) - X^{(i)}(\mathbf{w})| < \delta.$$

By the uniform continuity of g on K , it follows that for all $\mathbf{w} \in \text{supp}(\mathbf{w})$,

$$|g(X_n^{(1)}(\mathbf{w}), \dots, X_n^{(k)}(\mathbf{w})) - g(X^{(1)}(\mathbf{w}), \dots, X^{(k)}(\mathbf{w}))| < \epsilon.$$

Thus,

$$\sup_{\mathbf{w} \in \text{supp}(\mathbf{w})} |g(X_n^{(1)}(\mathbf{w}), \dots, X_n^{(k)}(\mathbf{w})) - g(X^{(1)}(\mathbf{w}), \dots, X^{(k)}(\mathbf{w}))| < \epsilon.$$

Combining the above, we have, as $n \rightarrow \infty$,

$$P\left(\sup_{\mathbf{w} \in \text{supp}(\mathbf{w})} |g(X_n^{(1)}(\mathbf{w}), \dots, X_n^{(k)}(\mathbf{w})) - g(X^{(1)}(\mathbf{w}), \dots, X^{(k)}(\mathbf{w}))| \geq \epsilon\right) \leq P(A_n) \rightarrow 0.$$

Therefore, as $n \rightarrow \infty$,

$$\sup_{\mathbf{w} \in \text{supp}(\mathbf{w})} |g(X_n^{(1)}(\mathbf{w}), \dots, X_n^{(k)}(\mathbf{w})) - g(X^{(1)}(\mathbf{w}), \dots, X^{(k)}(\mathbf{w}))| \xrightarrow{p} 0.$$

□

C.6.9 Proof of Theorem C.8

Proof of Theorem C.8. We establish the uniform validity of the bootstrap in two steps: first, we approximate the U-statistic with its Hájek projection to isolate the leading linear term; second, we apply the Giné-Zinn bootstrap central limit theorem to this linear term.

The Giné-Zinn framework is suitable here because it operates as a “transfer principle”: it establishes that if the function class indexing the original empirical process is uniformly Donsker, then the bootstrap process automatically converges to the same Gaussian limit. This general machinery is crucial for our setting because it implicitly handles the triangular array structure of the estimator without explicit verification of Lindeberg conditions.

Step 1: Bootstrap Hoeffding Decomposition The bootstrap BNN estimator $\tau_n^*(m)(\mathbf{w})$ is a U-statistic based on the bootstrap sample. We apply the Hoeffding decomposition to separate it into a leading linear term (an average of independent variables) and a remainder. Conditional on the original sample, we write:

$$\tau_n^*(m)(\mathbf{w}) - \mathbb{E}^*[\tau_n^*(m)(\mathbf{w})] = \frac{m}{n} \sum_{i=1}^n g_{1,n}^*(Z_i^*; \mathbf{w}) + R_n^*(\mathbf{w}),$$

where $\frac{m}{n} \sum_{i=1}^n g_{1,n}^*(Z_i^*; \mathbf{w}) =: L_n^*(\mathbf{w})$ is the linear projection term with kernel $g_{1,n}^*(z; \mathbf{w})$. The term $R_n^*(\mathbf{w})$ captures higher-order degenerate U-statistic terms.

Step 2: Weak Convergence of the Linear Term (Giné-Zinn) Consider the class of functions $\mathcal{F}_n = \{g_{1,n}(\cdot; \mathbf{w}) : \mathbf{w} \in \text{supp}(\mathbf{w})\}$. This class satisfies the bracketing entropy integral condition and has a uniformly bounded envelope, making it uniformly Donsker even as the distribution changes with n . Let $T_n^*(\mathbf{w}) = \frac{L_n^*(\mathbf{w})}{\sigma_n^*(\mathbf{w})}$ be the studentized linear process. The Giné-Zinn bootstrap CLT says that if the original process converges weakly to a tight Gaussian process $\mathbb{G}$, then the bootstrap process converges weakly to the same limit conditional on the sample. Thus: $T_n^* \rightsquigarrow \mathbb{G}$.

Step 3: Negligibility of the Remainder Since standard deviation $\sigma_n(\mathbf{w})$ scales as $O(\sqrt{m/n})$, the studentized remainder decays as:

$$\sup_{\mathbf{w} \in \text{supp}(\mathbf{w})} \left| \frac{R_n^*(\mathbf{w})}{\sigma_n^*(\mathbf{w})} \right| = O_{p^*} \left(\frac{m}{n} \right) = o_{p^*}(1),$$

given the assumption $m/n \rightarrow 0$. Thus, the asymptotic distribution of the BNN estimator is determined entirely by the linear term T_n^* .

Step 4: Uniform Convergence of CDF By Polya's Theorem, weak convergence to a continuous limit implies uniform convergence of the cumulative distribution functions. Therefore,

$$\sup_{\mathbf{w} \in \text{supp}(\mathbf{w})} \sup_{x \in \mathbb{R}} \left| \mathbb{P}_n^* \left(\frac{\tau_n^*(m)(\mathbf{w}) - \mathbb{E}^*[\tau_n^*(m)(\mathbf{w})]}{\sigma_n^*(\mathbf{w})} \leq x \right) - \Phi(x) \right| \xrightarrow{p} 0.$$

This completes the proof.

Remark: Simultaneous Confidence Bands While we establish the uniform validity of point-wise confidence intervals, constructing simultaneous confidence bands requires addressing the distribution of the supremum. Specifically, one must verify the anti-concentration property (Chernozhukov et al., 2015) to control the behavior of $\sup_{\mathbf{w}} |T_n(\mathbf{w})|$. The anti-concentration condition requires that for the supremum of the normalized process:

$$\sup_{t \in \mathbb{R}} \mathbb{P} \left(\left| \sup_{\mathbf{w} \in \text{supp}(\mathbf{w})} T_n(\mathbf{w}) \right| \in [t, t + \epsilon] \right) \leq A\epsilon^a,$$

for some constants $A, a > 0$ and all $\epsilon > 0$. Under our assumptions, this property is likely satisfied because: (i) the density $\nu(\cdot)$ is bounded away from zero (Assumption C.3) and the kernel is bounded (Assumption C.4), which ensures the limit process is tight and well-defined on the compact support; and (ii) the asymptotic variance is uniformly bounded away from zero, $\inf_{\mathbf{w}} \sigma^2(\mathbf{w}) \geq c > 0$. These conditions collectively ensure that the limit Gaussian process is non-degenerate and does not concentrate on any single value. While these properties support the validity of simultaneous inference, we do not provide a formal proof of this result in the current appendix. $\square$

C.6.10 Proof of Theorem C.9

Proof of Theorem C.9. We proceed in several steps.

Step 1: Uniform CLT for Original Sample From Theorem C.4, we have the uniform asymptotic normality of the original estimator: as $n \rightarrow \infty$, $m \rightarrow \infty$, and $m/n \rightarrow 0$,

$$\sup_{\mathbf{w} \in \text{supp}(\mathbf{w})} \sup_{x \in \mathbb{R}} \left| \mathbb{P} \left(\frac{\tau_n(m)(\mathbf{w}) - \mu(\mathbf{w}) - B(m)}{\sigma_n(\mathbf{w})} \leq x \right) - \Phi(x) \right| \rightarrow 0,$$

where $\Phi(x)$ is the cumulative distribution function of the standard normal distribution, $B(m)$ is the bias term as defined in Theorem C.1, and $\sup_{\mathbf{w}} \sigma_n^2(\mathbf{w}) = O(m/n)$ and $\inf_{\mathbf{w}} \sigma^2(\mathbf{w}) \geq c > 0$.

Step 2: Uniform CLT for Bootstrap Sample By Theorem C.8, we have established that the bootstrap distribution converges uniformly to the same normal limit as the original estimator. Specifically, conditional on the original sample:

$$\sup_{\mathbf{w} \in \text{supp}(\mathbf{w})} \sup_{x \in \mathbb{R}} \left| \mathbb{P}_n^* \left(\frac{\tau_n^*(m)(\mathbf{w}) - \mathbb{E}_{\mathbb{P}_n^*}[\tau_n^*(m)(\mathbf{w})]}{\sigma_n^*(\mathbf{w})} \leq x \right) - \Phi(x) \right| \xrightarrow{p} 0,$$

where $\mathbb{P}_n^*$ and $\mathbb{E}_{\mathbb{P}_n^*}$ are probability and expectation with respect to the bootstrap distribution, conditional on the original sample.

Step 3: Triangular Inequality Let $F_n^*(x, \mathbf{w})$ and $F_n(x, \mathbf{w})$ be the CDFs of the normalized bootstrap and original estimators:

$$F_n^*(x, \mathbf{w}) = \mathbb{P}_n^* \left(\frac{\tau_n^*(m)(\mathbf{w}) - \mathbb{E}_{\mathbb{P}_n^*}[\tau_n^*(m)(\mathbf{w})]}{\sigma_n^*(\mathbf{w})} \leq x \right),$$

$$F_n(x, \mathbf{w}) = \mathbb{P} \left(\frac{\tau_n(m)(\mathbf{w}) - \mathbb{E}[\tau_n(m)(\mathbf{w})]}{\sigma_n(\mathbf{w})} \leq x \right).$$

By the triangular inequality:

$$|F_n^*(x, \mathbf{w}) - F_n(x, \mathbf{w})| \leq |F_n^*(x, \mathbf{w}) - \Phi(x)| + |\Phi(x) - F_n(x, \mathbf{w})|.$$

Taking supremum over both x and $\mathbf{w}$:

$$\begin{aligned} & \sup_{\mathbf{w} \in \text{supp}(\mathbf{w})} \sup_{x \in \mathbb{R}} |F_n^*(x, \mathbf{w}) - F_n(x, \mathbf{w})| \\ & \leq \sup_{\mathbf{w} \in \text{supp}(\mathbf{w})} \sup_{x \in \mathbb{R}} |F_n^*(x, \mathbf{w}) - \Phi(x)| + \sup_{\mathbf{w} \in \text{supp}(\mathbf{w})} \sup_{x \in \mathbb{R}} |\Phi(x) - F_n(x, \mathbf{w})| \xrightarrow{p} 0, \end{aligned}$$

where the convergence follows from Steps 1 and 2. Therefore,

$$\sup_{\mathbf{w} \in \text{supp}(\mathbf{w})} \sup_{x \in \mathbb{R}} |F_n^*(x, \mathbf{w}) - F_n(x, \mathbf{w})| \xrightarrow{p} 0,$$

which establishes the uniform validity of the bootstrap procedure. $\square$

C.6.11 Proof of Lemma 2

Proof. By the definition of Hadamard differentiability, the difference in the transformed estimators can be approximated by the derivative applied to the difference in the inputs. Let $\Delta_n^*(\mathbf{w}) = r_n(\hat{\theta}_n^*(\mathbf{w}) - \hat{\theta}_n(\mathbf{w}))$. We can write:

$$r_n(g(\hat{\theta}_n^*(\mathbf{w})) - g(\hat{\theta}_n(\mathbf{w}))) = g'_{\theta}(\mathbf{w})(\Delta_n^*(\mathbf{w})) + R_n^*(\mathbf{w})$$

where the remainder term $R_n^*(\mathbf{w})$ satisfies $\sup_w |R_n^*(\mathbf{w})| = o_{p^*}(1)$.

By Assumption 2, the input process $\Delta_n^*(\cdot)$ converges weakly to the Gaussian process $\mathbb{G}(\cdot)$. Since the derivative map $h \mapsto g'_\theta(h)$ is a continuous linear operator, the Continuous Mapping Theorem implies that:

$$g'_{\theta(\cdot)}(\Delta_n^*(\cdot)) \rightsquigarrow g'_{\theta(\cdot)}(\mathbb{G}(\cdot)) \quad \text{in } \ell^\infty(\text{supp}(\mathbf{w})).$$

The limit process $\mathbb{Z}(\mathbf{w}) := g'_{\theta}(\mathbf{w})(\mathbb{G}(\mathbf{w}))$ is a Gaussian process because it is a linear transformation of a Gaussian process.

For any fixed $\mathbf{w}$, the limit random variable $\mathbb{Z}(\mathbf{w})$ follows a normal distribution with variance $\sigma_g^2(\mathbf{w}) = \text{Var}(g'_{\theta}(\mathbf{w})(\mathbb{G}(\mathbf{w})))$. To establish the validity of the studentized statistic, we rely on the consistency of the bootstrap variance estimator. Since the underlying process is Donsker and bounded, the bootstrap consistency of the distribution implies the consistency of the second moments. Thus,

$$\frac{r_n}{\hat{\sigma}_g(\mathbf{w})} \xrightarrow{p^*} \frac{1}{\sigma_g(\mathbf{w})} \quad \text{uniformly in } \mathbf{w}.$$

We now apply the Slutsky's theorem. Since the numerator converges weakly to $\mathbb{Z}(\mathbf{w})$ and the denominator converges in probability to the constant $\sigma_g(\mathbf{w})$, the ratio converges to the standardized limit:

$$r_n \frac{g(\hat{\theta}_n^*(\mathbf{w})) - g(\hat{\theta}_n(\mathbf{w}))}{\hat{\sigma}_g(\mathbf{w})} \rightsquigarrow \frac{\mathbb{Z}(\mathbf{w})}{\sigma_g(\mathbf{w})} \sim N(0, 1).$$

Since the limit distribution is the standard normal $\Phi(x)$, which is continuous and does not depend on $\mathbf{w}$, Polyá's Theorem implies that the convergence of the CDF is uniform over $x \in \mathbb{R}$ and $\mathbf{w} \in \text{supp}(\mathbf{w})$. $\square$

C.6.12 Proof of Theorem C.10

Proof. This result follows from the Bootstrap Continuous Mapping Theorem (Lemma 2). The finite difference estimator is a linear (and thus Hadamard differentiable) transformation of the stochastic process $\tau_n(m)(\cdot)$. Since Theorem C.8 establishes the weak convergence of the underlying process, and the bandwidth condition $m^{4/d}h_n \rightarrow \infty$ ensures the bias is asymptotically negligible (as shown in Theorem C.5), the conditions of Lemma 2 are satisfied. The uniform validity of the bootstrap derivative estimator follows immediately. $\square$

C.6.13 Proof of Theorem C.11

Proof. This result follows from the Bootstrap Continuous Mapping Theorem (Lemma 2) applied to the vector of estimators.

Let $\mathbf{T}_n(\mathbf{w})$ denote the vector stacking all k_1 level estimators and k_2 derivative estimators. We have previously established that the bootstrap empirical process for each component converges weakly to a tight Gaussian process (Theorems C.8 and C.10). To establish joint convergence, we observe that these estimators are functionals of the same underlying empirical measure. Since the Donsker property is preserved under finite unions, the joint empirical process converges weakly on the product space. Thus, we obtain the joint weak convergence of the vector process:

$$r_n(\mathbf{T}_n^*(\cdot) - \mathbf{T}_n(\cdot)) \rightsquigarrow \mathbb{G}_{\text{joint}}(\cdot) \quad \text{in } \ell^\infty(\text{supp}(\mathbf{w}))^{k_1+k_2},$$

where $\mathbb{G}_{\text{joint}}$ characterizes the joint asymptotic distribution and covariance structure.

The transformation g is Hadamard differentiable by assumption. Since the input process converges uniformly, the conditions of Lemma 2 are satisfied. The lemma implies that the bootstrap distribution of the transformed estimator converges uniformly to the normal distribution determined by the linear approximation. $\square$

D Simulation Details and Additional Results

D.1 DGP Setups

Random-coefficient Logit Models We start with the random-coefficient logit model, a framework often used in empirical demand analysis (Berry et al., 1995). The DGP is that each consumer has a logit demand and chooses among a fixed set of alternatives. Preferences are fixed within each consumer and might differ across consumers. Similar DGPs are used to characterize micro-level choice data (Rossi et al., 1996). A special case is logit demand (McFadden, 1973; Berry, 1994), in which consumers are homogeneous apart from their utility shocks.

Setup. Let ι index consumers. Let $j = 1, 2, \dots, J$ index a fixed set of products. And let t index markets (t can also be interpreted as periods). Consumer ι buys at most one product out of the choice set and one unit of that product. If she buys product j , she derives utility

$$u_{\iota jt} = \beta_{\iota j} + \alpha_{\iota} p_{jt} + \xi_{jt} + \epsilon_{\iota jt} ,$$

where $\beta_{\iota j}$ characterizes product j 's match value to consumer ι , α_{ι} is the consumer's price sensitivity, ξ_{jt} is the product's unobserved characteristics in market t following a uniform distribution between $[-0.5, 0.5)$, and $\epsilon_{\iota jt}$ is a type-I extreme value utility shock. If the consumer chooses the outside good, she gets $u_{\iota 0t} = \epsilon_{\iota 0t}$. In this simulation, we model product-consumer match value $\beta_{\iota j}$ for simplicity. One could instead characterize preferences over observed product characteristics.

At the individual level, this model gives a logit choice probability, which has a closed-form derivative on price. Specifically, the probability of the consumer choosing j is

$$s_{\iota jt} = \frac{\exp(\beta_{\iota j} + \alpha_{\iota} p_{jt} + \xi_{jt})}{1 + \sum_{k=1}^J \exp(\beta_{\iota k} + \alpha_{\iota} p_{kt} + \xi_{kt})} .$$

And the derivative of individual choice probabilities on own price and other products' prices are

$$\frac{\partial s_{\iota jt}}{\partial p_{kt}} = \begin{cases} \alpha_{\iota} s_{\iota jt} (1 - s_{\iota jt}) & \text{if } j = k, \\ -\alpha_{\iota} s_{\iota jt} s_{\iota kt} & \text{if } j \neq k. \end{cases}$$

Researchers do not see individual-level choice probabilities. Instead, they observe the total sales quantity for each product in each market. Denote the vector of consumer hetero-

geneity $\Theta_\iota = (\beta_{\iota 1}, \dots, \beta_{\iota J}, \alpha_\iota)$. In the model, sales quantities are the sum of choice probabilities across all individuals:

$$s_{jt} = \mathcal{M} \int_\iota s_{\iota jt} dF(\Theta_\iota),$$

where constant $\mathcal{M}$ is the number of consumers in the market and $F(\Theta_\iota)$ is the consumer preference distribution. The derivative of sales quantity is

$$\frac{\partial s_{jt}}{\partial p_{kt}} = \mathcal{M} \int_\iota \frac{\partial s_{\iota jt}}{\partial p_{kt}} dF(\Theta_\iota).$$

We have not yet specified the distribution of consumer preferences, $F(\Theta_\iota)$. We consider three cases for this distribution. In the first case, consumer preferences are homogeneous. This case reduces to the logit model. In our simulations, we assume β equals 0.4 for all products and $\alpha = -3$.

In the second case, consumer preference parameters follow independent normal distributions. This specification is common in empirical applications of the random-coefficient demand model. Specifically, β and α follow independent normal distributions with means of 0 and -3 and standard deviations of 0.8 and 0.5, respectively.

The third specification further allows for the correlation between demand intercepts and random coefficients. Suppose log income loginc_ι drives consumers' demand for product quality (part of the intercept) and price sensitivities. We have $\beta_{\iota j} = b_{0j} + b_{1j}\text{loginc}_\iota$ and $\alpha_\iota = a_0 + a_1\text{loginc}_\iota + a_2\nu_\iota$, where ν follows a standard normal distribution. In our simulations, we take loginc_ι from a standard normal distribution. In addition, we assume $\mathbf{b}_0 = \mathbf{0}$ and $\mathbf{b}_1 = \mathbf{0.8}$ for all products, $a_0 = -3$, and $a_1 = a_2 = 0.5$.

We allow prices to be endogenous to unobserved product characteristics, ξ_{jt} . For all three DGPs, we specify a reduced-form selection equation:

$$\mathbf{p}_t = 0.5 \mathbf{1} + 0.5 \mathbf{z}_t + 0.5 \xi_t + 0.5 \mathbf{e}_t,$$

where prices are functions of (own) unobserved characteristics ξ , instruments z , and exogenous shocks e . In addition to ξ , the other two components z and e also follow uniform distributions in $[-0.5, 0.5]$. The inclusion of e reduces the power of price instruments and mimics realistic settings. Although prices in the baseline case are determined by a reduced-form process, we demonstrate in robustness checks that our estimator performs well when the prices are generated by a Bertrand-Nash equilibrium. We do not adopt structural, equilibrium prices in the baseline case because computing such equilibrium prices takes a

significant amount of time. This computation becomes infeasible for very large samples.

Demand when some products are complements. The random-coefficient logit model assumes that products are substitutes for each other. In reality, many products or services are complements (Smith et al., 2019; Compiani and Smith, 2021), in which case, the random-coefficient logit model will not estimate the correct cross elasticities. In addition, standard parametric aggregate-data demand models do not have a good way to accommodate complements. We now examine how our nonparametric demand estimator recovers own and cross elasticities in markets of substitutes and complements.

We follow Gentzkow (2007) and set up a DGP where some products are substitutes and others are complements. We assume that when an individual ι consumes product j at market t , she enjoys an indirect utility

$$u_{jt} = \delta - \alpha p_{jt} + \xi_{jt} + \epsilon_{jt}.$$

where all subscripts ι have been omitted since individuals are also identical as in Gentzkow (2007). The unobserved disturbance ϵ_{jt} follows the Type-1 extreme value distribution and is independent and identically distributed across products and markets. In addition, there is an outside option of no purchase with $u_{0t} = \epsilon_{0t}$.

This model allows the existence of a bundle. To be specific, we assume that one unit of Product 1 and one unit of Product 2 can be purchased as a bundle, and for the purchase of this bundle, the indirect utility is

$$u_{bt} = \delta - \alpha (p_{1t} + p_{2t}) + \gamma + u_{1t} + u_{2t} + \epsilon_{bt},$$

where γ is the extra utility when purchasing Product 1 and Product 2 as bundles. Gentzkow shows that when γ is positive enough, Product 1 and Product 2 can be complements. In other words, the extra degree of complementarity overcomes the implied substitution from the logit structure.

Our simulation introduces 20,000 markets with γ set to be constant 4. The bottom panel of Figure 1 shows our estimated point-wise own and cross-price elasticities, together with their closed-form counterparts in the true model. The closed-form derivations of the own and cross-price elasticities do not enjoy the clean forms as in the multinomial logit model. They vary across products and depend on the specification of the bundle composition, extra utility γ , and the preferences.

Demand when consumers can purchase multiple products and units. So far, all DGPs assume that consumers can only choose one product and purchase exactly one unit. The multiple discrete choice literature (Hendel, 1999; Kim et al., 2002; Dubé, 2004; Chan, 2006; Bhat, 2008) has shown that consumers sometimes purchase multiple products and in different quantities and has demonstrated that such variety/quantity behavior might lead to a different demand function.

We follow Kim et al. (2002), who model consumers as having decreasing marginal utility on each product variety and solving an inequality constrained optimization problem when making product choices. The consumer has a random utility with a translated CES utility function with product-specific parameters governing marginal utility and satiation. She maximizes this utility function subject to nonnegative consumption constraints and budget constraints. As a consequence, variety and quantity are allowed.

In this simulation, we assume consumer ι at market t faces the following problem

$$\begin{aligned} \max_{x_{\iota 1t}, x_{\iota 2t}, x_{\iota 3t}} \quad & U(x_{\iota 1t}, x_{\iota 2t}, x_{\iota 3t}) = \sum_{k=1}^3 \chi_k \frac{\gamma_k}{\alpha_k} \left\{ \left(\frac{x_{\iota kt}}{\gamma_k} + 1 \right)^{\alpha_k} - 1 \right\} \\ \text{s.t.} \quad & x_{\iota 1t}, x_{\iota 2t}, x_{\iota 3t} \geq 0, \\ & p_{1t}x_{\iota 1t} + p_{2t}x_{\iota 2t} + p_{3t}x_{\iota 3t} \leq e_{\iota}. \end{aligned}$$

where $\alpha_k, \gamma_k, \chi_k$ are preference parameters for each of the 3 existing products and are assumed to be constant across markets and individuals. e_{ι} is individual specific budget, p_{jt} are price for product j in market t .

Solving the above optimization problem with inequality constraints is computationally intensive, and the computation burden grows exponentially with the number of products J . Therefore, in our simulations, we limit $J = 3$.

D.2 Literature-Calibrated DGPs

The baseline DGPs in Section D.1 use stylized parameter values. To further assess our estimator under realistic demand primitives, we calibrate the random-coefficient logit framework to published parameter estimates from three studies: Wollmann (2018), Nevo (2001), and Berry et al. (1995). Direct replication of these papers would also require changing the number of products J . The original studies involve many more products. By contrast, we keep $J = 4$ fixed to stay consistent with our other simulations, leaving the exploration of higher J as a separate exercise (Appendix Table A2, Panels B and C). To better mimic these DGPs while keeping $J = 4$, we adopt a calibration approach that preserves the key preference parameters governing price sensitivity and consumer heterogeneity from each source paper, while adjusting the remaining parameters to produce a DGP whose equilibrium outcomes—elasticity magnitudes, market share distributions, and price ranges—are consistent with those reported in the original studies.

D.2.1 Model specification

All three calibrations share the same utility specification:

$$u_{\iota jt} = \gamma_{\iota} + \alpha_{\iota} p_{jt} + \xi_{jt} + \epsilon_{\iota jt},$$

where $\gamma_{\iota} = \gamma_0 + \gamma_1 \log \text{inc}_{\iota} + \gamma_2 \nu_{1\iota}$ is the consumer-specific intercept, $\alpha_{\iota} = \alpha_0 + \alpha_1 \log \text{inc}_{\iota} + \alpha_2 \nu_{2\iota}$ is the consumer-specific price coefficient, ν_1, ν_2 are drawn independently from the standard normal distribution, $\log \text{inc}$ from a log-normal income distribution, ξ_{jt} is the product-market unobservable (uniform on $[-0.5, 0.5]$), and $\epsilon_{\iota jt}$ is a type-I extreme value shock. The product-level unobservable includes a fixed component $\bar{\xi}_j$ that differentiates products in quality. Prices follow the reduced-form equation $p_{jt} = \bar{p}_j + c_z z_{jt} + c_{\xi} \xi_{jt}$, where z_{jt} is a cost-side instrument and c_{ξ} governs the degree of price endogeneity.

D.2.2 Calibration procedure

For each source paper, we partition the parameters into two groups. The first group consists of parameters set based on published estimates: the mean price coefficient α_0 , income interactions α_1 and γ_1 , and random-coefficient standard deviations α_2 and γ_2 . These parameters govern the price sensitivity and preference heterogeneity that are central to the elasticity patterns we aim to reproduce.

The second group consists of parameters that we calibrate. The mean intercept γ_0 is the primary calibration target: we select it via binary search so that the simulated mean absolute own-price elasticity matches the value reported in the source paper. This is necessary because γ_0 controls the level of inside-good market shares, which in turn affects the magnitude of price elasticities through the logit structure. The product quality spread $\bar{\xi}_j$ is chosen to generate differentiated market shares across the four products, producing a realistic distribution rather than symmetric shares. The price-generating parameters—base price levels $\bar{p}_j$, instrument scale c_z , and endogeneity scale c_ξ —are set to produce price distributions in ranges consistent with each source paper’s empirical setting.

D.2.3 Calibration details for each DGP

The first calibration draws on the logit specification in [Wollmann \(2018\)](#), who estimates demand for commercial trucks with a price coefficient of $\alpha_0 = -0.44$ per \$1,000 and mean own-price elasticities around -8.5 . All random-coefficient and income-interaction parameters are set to zero, yielding a standard logit model. We set base prices at \$15,000–\$35,000 to approximate the price range for heavy trucks, and use a wide quality spread ($\bar{\xi}_j \in [-14, 14]$) to generate the substantial market share differentiation observed in that industry.

The second calibration is based on [Nevo \(2001\)](#), using the “best practices” estimates from [Conlon and Gortmaker \(2020\)](#) (Table 7). This specification features a large mean price coefficient ($\alpha_0 = -27.5$, reflecting prices in dollars per serving) with substantial heterogeneity from both income interactions ($\alpha_1 = 0.8$) and an independent random coefficient ($\alpha_2 = 2.9$). The target mean elasticity is -3.7 . Base prices are set near \$0.15 per serving, consistent with the ready-to-eat cereal market.

The third calibration follows [Berry et al. \(1995\)](#), again using the replication in [Conlon and Gortmaker \(2020\)](#) (Table 8). The original BLP specification uses a nonlinear price term $\alpha \ln(y_t - p_{jt})$; we approximate this with a linear specification $\alpha_0 = -0.30$ (roughly $\alpha/\bar{y}$) and a small income interaction $\alpha_1 = 0.05$. The constant has a large random coefficient ($\gamma_2 = 3.0$), generating substantial unobserved preference heterogeneity. The target mean elasticity is -3.5 , and base prices are set near \$15,000 to match the automobile market.

D.2.4 Validation

Table [A1](#) reports the complete parameterization for each calibration and the achieved simulation outcomes. Panel D confirms that the calibration procedure successfully matches the target elasticities: the achieved mean absolute own-price elasticities are 8.50, 3.68, and 3.47

against targets of 8.5, 3.7, and 3.5, respectively. The resulting market structures are also realistic. The Wollmann calibration produces an outside-good share of 0.19, with inside shares ranging from 0.01 to 0.41, reflecting the concentrated market for heavy trucks. The Nevo calibration yields a large outside-good share of 0.72, consistent with the many unmodeled cereal brands outside the top four. The BLP calibration produces an outside-good share of 0.22, with inside shares between 0.06 and 0.36, reflecting the differentiated automobile market.

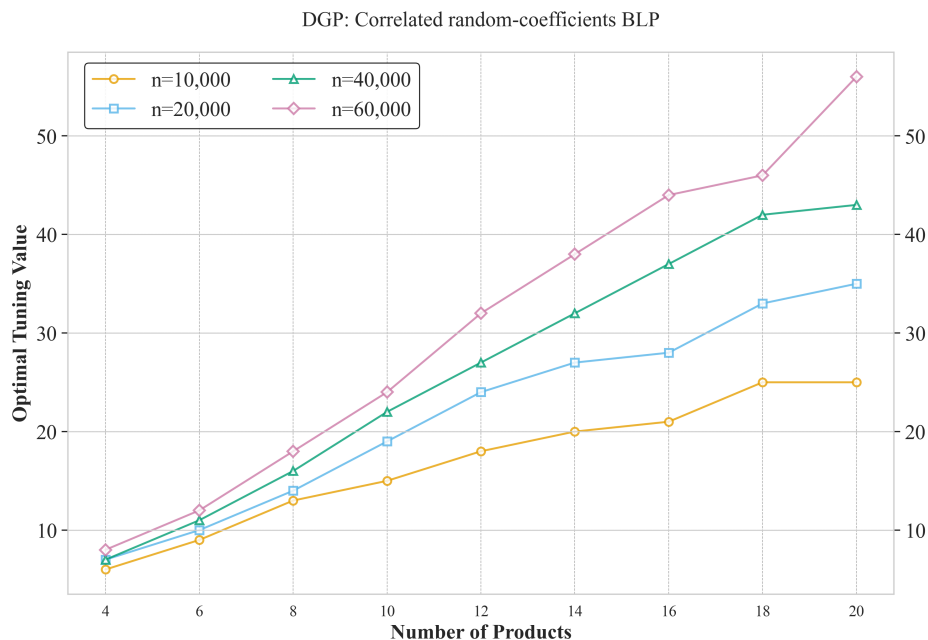
Table A1: Parameter Values for Literature-Calibrated DGPs

	Wollmann (2018) Logit	Nevo (2001) RC Logit	BLP (1995) RC Logit
<i>Panel A: Demand parameters based on source papers</i>			
α_0 (price mean)	-0.44	-27.5	-0.30
α_1 (price $\times$ income)	0	0.8	0.05
α_2 (price SD)	0	2.9	0
γ_1 (constant $\times$ income)	0	0.5	0.5
γ_2 (constant SD)	0	0.2	3.0
<i>Panel B: Calibrated parameters</i>			
γ_0 (constant mean)	10.76	0.24	5.51
$\bar{\xi}_j$ (quality spread)	$[-14, -4, 4, 14]$	$[6, 2, -2, -6]$	$[2, 0.7, -0.7, -2]$
<i>Panel C: Price-generating parameters</i>			
$\bar{p}_j$ (base price)	$[19, 23, 28, 35]$	0.15	15.0
c_z (instrument scale)	3.0	0.05	5.0
c_ξ (endogeneity scale)	0.5	0.01	1.0
<i>Panel D: Calibration targets and achieved outcomes</i>			
Target mean $ \varepsilon $	8.5	3.7	3.5
Achieved mean $ \varepsilon $	8.50	3.68	3.47
Outside-good share	0.19	0.72	0.22
Inside share range	$[0.01, 0.41]$	$[0.002, 0.22]$	$[0.06, 0.36]$
Price range	$[15.7, 38.5]$	$[0.12, 0.18]$	$[14.0, 16.5]$

Note: This table reports the full parameterization and achieved outcomes for three literature-calibrated DGPs used in Table 2. Panel A: preference parameters set directly from published estimates. Panel B: γ_0 obtained via binary search to match the target mean absolute own-price elasticity; $\bar{\xi}_j$ chosen to produce differentiated market shares. Panel C: reduced-form price equation parameters governing price levels, instrument-driven variation, and endogeneity. Panel D: simulation outcomes averaged over 10 Monte Carlo replications with $T = 50,000$ and $J = 4$ products. See text for details on the calibration procedure and source-specific adaptations.

D.3 Additional Simulation Results

Figure A2: Optimal tuning parameter choice



Note: The figure shows how optimal tuning parameters vary with both the number of products and sample size in a correlated random-coefficient BLP model. Lines with matching markers connect optimal subsampling sizes (see the legend) with identical sample sizes but different numbers of products. The horizontal axis represents the number of products, while the vertical axis indicates the optimal subsampling size.

Table A2: Robustness checks

Specification	Elasticity	True Value	Mean Estimate	Bias	Mean SE	Std. Dev.
<i>Panel A: Varying sample size ($J = 4$ products)</i>						
$T = 5,000$	own	-1.331	-1.380	-0.049	0.183	0.208
	cross	0.204	0.214	0.010	0.175	0.189
$T = 20,000$	own	-1.337	-1.361	-0.024	0.094	0.102
	cross	0.205	0.212	0.007	0.086	0.083
$T = 80,000$	own	-1.331	-1.379	-0.048	0.046	0.047
	cross	0.203	0.205	0.002	0.043	0.045

Continued on next page

Table A2 (continued)

Specification	Elasticity	True Value	Mean Estimate	Bias	Mean SE	Std. Dev.
$T = 320,000$	own	-1.332	-1.360	-0.028	0.023	0.028
	cross	0.203	0.201	-0.002	0.022	0.023
<i>Panel B: Varying number of products ($T = 320,000$)</i>						
$J = 4$	own	-1.334	-1.326	0.008	0.040	0.046
	cross	0.205	0.203	-0.002	0.037	0.035
$J = 8$	own	-1.394	-1.405	-0.011	0.083	0.082
	cross	0.128	0.132	0.004	0.070	0.087
$J = 12$	own	-1.428	-1.592	-0.164	0.118	0.111
	cross	0.095	0.096	0.001	0.094	0.083
$J = 16$	own	-1.441	-1.896	-0.455	0.152	0.150
	cross	0.075	0.081	0.006	0.115	0.104
<i>Panel C: Varying number of products ($T = 5,120,000$)</i>						
$J = 12$	own	-1.387	-1.368	0.019	0.123	0.135
	cross	0.113	0.092	-0.021	0.107	0.093
$J = 16$	own	-1.411	-1.402	0.009	0.158	0.150
	cross	0.089	0.087	-0.002	0.133	0.131
$J = 20$	own	-1.427	-1.499	-0.072	0.188	0.170
	cross	0.073	0.073	0.000	0.157	0.172
$J = 24$	own	-1.439	-1.615	-0.176	0.221	0.188
	cross	0.061	0.078	0.017	0.176	0.189

Note: This table presents robustness checks for point-wise price elasticity estimates across different sampling specifications. Panel A varies the sample size while holding the number of products constant. Panels B and C maintain a fixed sample size while varying the number of products. The analysis employs tuning parameters of 7, 30, and 480. For each panel, columns report: the number of products or sample size (depending on panel), the type of elasticity (own-price or cross-price), true elasticity value, mean estimated elasticity across simulations, estimation bias (mean estimate minus true value), mean standard error of the estimates, and the standard deviation of estimates across simulations.

Table A3: Mean Squared Error Minimizing Tuning Parameter Choice (Extended)

Panel A: Discrete-choice Models for Cross Elasticity

Model	Obs.	Number of Products								
		4	6	8	10	12	14	16	18	20
Logit	10,000	5	5	5	4	4	3	3	3	3
	20,000	5	6	6	5	5	4	4	4	3
	40,000	5	7	7	7	6	6	5	5	4
	60,000	6	7	7	7	7	7	6	6	5
Independent RC Logit	10,000	5	5	5	4	4	4	3	3	3
	20,000	5	6	6	5	5	5	4	4	4
	40,000	6	6	7	7	6	6	5	5	5
	60,000	6	7	7	8	7	7	6	6	6
Correlated RC Logit	10,000	5	5	5	4	4	4	3	3	3
	20,000	5	6	6	6	5	5	4	4	4
	40,000	6	7	7	7	7	7	6	5	5
	60,000	6	7	8	7	7	7	6	5	6

Panel B: Reduced-form Polynomial Models

Model	Obs.	Number of Products								
		4	6	8	10	12	14	16	18	20
Reduced-form linear	10,000	4	5	6	6	7	8	8	9	10
	20,000	5	6	7	7	8	9	10	11	12
	40,000	5	6	7	8	10	11	12	13	14
	60,000	5	7	8	9	10	12	13	14	16
Reduced-form quadratic	10,000	4	5	5	6	7	8	8	9	10
	20,000	4	5	6	7	8	9	10	11	12
	40,000	4	5	7	8	9	11	12	13	14
	60,000	4	5	7	8	9	11	13	15	15

Note: Panel A focuses on cross-price elasticities using three baseline random-coefficients models as data-generating processes, while Panel B examines own-price elasticities using reduced-form polynomial models. Both panels examine optimal subsampling sizes across varying sample sizes (10,000 to 60,000 observations) and product counts (4 to 20 products in increments of 2). For each configuration, we estimate elasticities at the mean price vector using subsampling sizes (m) from 2 to 100 for 4, 8, 16, and 20 products and from 2 to 60 for the remaining product counts in Panel A, and from 2 to 30 in Panel B (all reported optima are interior to these ranges), performing 1,000 Monte Carlo simulations per size. The optimal subsampling size is determined by minimizing the Mean Squared Error (MSE) across the 1,000 simulations.

Table A4: Point-wise Price Elasticity Estimation for extra primitive parameter values

Panel A Primitives: $b_1 = 4, a_0 = -1.2, a_1 = 0.3, a_2 = 0.8$

Model	Elasticity	True Value	Mean Estimate	Bias	Mean SE	Std. Dev.
Correlated RC Logit	own	-0.457	-0.462	-0.005	0.058	0.063
	cross	0.152	0.146	-0.006	0.057	0.050

Panel B Primitives: $b_1 = 1.2, a_0 = -4, a_1 = 0.8, a_2 = 0.3$

Model	Elasticity	True Value	Mean Estimate	Bias	Mean SE	Std. Dev.
Correlated RC Logit	own	-1.504	-1.527	-0.022	0.108	0.097
	cross	0.501	0.510	0.008	0.098	0.098

Note: This table reports estimation results for the random-coefficient logit model under two alternative parameter specifications. Elasticities are evaluated at the mean price vector, averaged across 50 Monte Carlo simulations. Both panels utilize $J = 4$ products and a sample size of $T = 20,000$, with subsampling sizes of $m = 7$ (own-price) and $m = 5$ (cross-price). Columns report: elasticity type, true value, mean point estimate, bias, mean bootstrap standard error, and the standard deviation of estimates across simulations.

E Supplementary Empirical Results

Table A5: Distribution of Zone-Level Price Elasticity Estimates

	Mean	Min	5th	10th	25th	50th	75th	90th	95th	Max
97 zones	0.91	-42.04	-7.36	-4.47	-2.83	-1.47	0.20	3.17	4.62	221.87
73 zones	-1.89	-42.04	-8.30	-4.69	-2.98	-1.71	-0.85	0.89	3.48	37.08

Notes: The 73-zone subsample excludes zones with fewer than 50,000 observations and zones where more than 65% of market cells have zero transaction volume. The standard deviation of the estimates is 23.86 for the full sample (97 zones) and 7.77 for the restricted sample (73 zones).

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